

On Causality as a Role in Protein Knowledge Graphs: Addressing Without Storage, Edges Without Curation

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Abstract

A protein knowledge graph is normally built by storing things: nodes for proteins, edges for relations, and curated assertions binding the two. We show that neither needs to be stored. Nodes are generated by an addressing scheme that assigns each biomolecular object a positional name computed directly from its physical observables, so that identity is a *computation* rather than a lookup; edges are generated by a propagation rule on a weighted contact graph, so that relations are *derived* rather than asserted. The two constructions turn out to be one construction. We derive, from a single order condition on the contact structure — that not every object can be a medium for every other — a strictly positive *contact floor* $\beta > 0$, and show that individuation must proceed by cutting rather than by labelling. Identity is then the minimum cut separating an object from its medium: an invariant, never a point, and never a name. Independently, we derive a three-coordinate entropy space $\mathcal{S} = [0, 1]^3$ for biomolecular objects, prove that base-three positional encoding is the unique base for which digit depth and coordinate resolution advance in lockstep, and obtain four query operations — identify, similar, predict, react — each running in time linear in address depth with storage independent of how many objects the scheme can answer about. The central result (Theorems 9.3, 10.3 and 11.1) is that entropic separation *is* the contact weighting the propagation framework otherwise takes as given, that minimum cuts under this weighting are realised on subtrie boundaries even though minimum cut is in general not an ultrametric, and consequently that the address catalogue and the propagation table are the same object: the catalogue is the table at rest, the table is the

catalogue propagated. A knowledge graph built this way answers admissibility questions — *could this have become that, and by what accountable route* — which retrieval over stored triples cannot express, and it does so with $O(1)$ storage. We work the construction through Cu/Zn superoxide dismutase and its familial-ALS variants, where a label-based graph records an association and this one computes a propagation account for it.

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9.1 What has to be derived	19	Every large protein resource in current use is a container. The Protein Data Bank stores structures (Berman et al., 2000; Burley et al., 2023); UniProt stores entries (The UniProt Consortium, 2023); Reactome stores reactions (Milacic et al., 2024); the Gene Ontology stores terms and the edges between them (Ashburner et al., 2000; The Gene Ontology Consortium, 2023); a knowledge graph in the semantic-web sense stores triples (Hogan et al., 2021; Unni et al., 2022). The engineering that follows from this — persistent identifiers, cross reference tables, curation pipelines, ontology alignment, versioning policy — is not incidental overhead. It is the direct consequence of having decided, before any biology enters, that knowledge about proteins is a set of records to be kept.	
9.2 Fixing the profile	19	The costs are well documented and are not, at this point, controversial. Identifiers decay and must be maintained (McMurry et al., 2017; Wren, 2008). Annotation carries bias that propagates through downstream inference (Gaudet and Dessimoz, 2017). Cross-resource mappings are curated by hand and rot when either side changes. The FAIR programme (Wilkinson	
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et al., 2016) is in substantial part a response to the container assumption: if knowledge is stored, then findability, accessibility, interoperability and reuse are properties of the storage, and each must be separately engineered and separately maintained.

This paper asks a different question. Not *how do we curate the container better*, but: what would a protein knowledge graph look like if it stored nothing at all?

The answer requires two things that a container does not supply.

Nodes without records. A node in a stored graph is a record with a label — an accession, a PDB identifier, a term ID. The label is the identity. But a label is an arbitrary assignment: two curators may attach different labels to the same object, and the same label to different objects, and nothing internal to the graph detects it. We show in Theorem 3.5 that no label carries invariant content, which is exactly why mappings between resources must be curated rather than computed. The alternative is to make identity a *computation on the object itself*: a positional address derived from measurable quantities, such that two objects have the same address if and only if they are indistinguishable at the resolution of the address. Nothing is stored, because the address is recomputed on demand from the object. This is developed in Sections 6 and 8.

Edges without assertions. An edge in a stored graph is an assertion by a curator that a relation holds. Its warrant is external to the graph. The alternative is to derive relations from a propagation rule: given a weighted contact structure, an edge is not an assertion but the statement that a propagation from one region to another is *accountable* — that it can be tracked without loss of account. This requires a notion of separation cost, a floor below which separation cannot fall, and an alignment functional bounded by that floor. All three are derived in Sections 2 and 4.

The claim. These two constructions — addressing without records, propagation without assertions — are not two halves of a hybrid system. They are the same construction approached from opposite ends. The addressing scheme, developed for identification, turns out

to *supply* precisely the contact weighting that the propagation framework requires and cannot itself produce; the propagation framework turns out to *supply* precisely the notion of relation that the addressing scheme lacks. Theorem 11.1 makes this exact: the address catalogue and the propagation table are the same object, the first at rest and the second in motion.

The consequence for practice is a query semantics that a stored graph cannot express. A retrieval query asks *what is recorded*. An admissibility query asks *what could have happened, and by what accountable route* — and returns an equivalence class of propagations rather than a set of tuples. Section 13 develops this and Section 15 works it through a concrete disease system.

What is derived here. This paper is self-contained. The contact floor, individuation by negation, cut-valued identity, the alignment functional, the representation freedoms, the propagation table, the entropy coordinates, the naturalness of base three, the addressing hierarchy, the four query operations, and the storage bound are all proved below from stated axioms. External citations are to the empirical record and to standard mathematics, never to the constructions themselves.

Road map. Sections 2 to 5 develop the propagation side: contact, floor, identity as cut, alignment, accountability, and the table. Sections 6 to 8 develop the addressing side: entropy coordinates, ternary encoding, hierarchical addresses, and the four operations. Sections 9 and 10 join them: entropic separation is derived as the contact weighting, minimum cuts are shown to be realised on subtrie boundaries, and the catalogue and the table are proved identical. Sections 13, 15 and 16 treat query semantics, a worked disease instance, and the boundaries of the construction.

Part I. Propagation without assertions

2 Contact, medium, and the floor

We begin with the weakest structure that can carry a notion of “related to” without presupposing what the relation means.

2.1 The contact structure

Definition 2.1 (Contact graph). A *contact graph* is a finite weighted graph $\Gamma = (V, E, w)$ where V is a finite set of *items*, $E \subseteq \binom{V}{2}$ is a set of unordered pairs, and $w: E \rightarrow (0, \infty)$ assigns each pair a positive *contact weight*. We write $w(F) := \sum_{e \in F} w(e)$ for $F \subseteq E$, and $\Omega := w(E)$ for the total weight.

No metric is assumed on V , no measure, no probability. The weight $w(\{u, v\})$ is to be read as the cost of separating u from v , not as a distance between them; the distinction matters, because separation costs add along a cut while distances do not.

Definition 2.2 (Cut). For $S \subseteq V$, the *cut* $\partial(S)$ is the set of edges with exactly one endpoint in S :

$$\partial(S) := \{ \{u, v\} \in E : |\{u, v\} \cap S| = 1 \}.$$

We write $S^c := V \setminus S$, and note $\partial(S) = \partial(S^c)$.

Definition 2.3 (Medium). A contact graph is *grounded* if it carries a distinguished item $\mathbf{m} \in V$, the *medium*, such that every $v \in V$ is connected to $\mathbf{m}$ in Γ .

The medium is whatever an item is separated *from*: in solution, the solvent; in a cell, the cellular milieu. It is a structural slot, not an assumption about chemistry. Its role is to make “separated” a two-place relation with one place fixed, which is what makes separation cost well defined.

Definition 2.4 (Separation cost). For $v \in V \setminus \{\mathbf{m}\}$, the *separation cost* is

$$\sigma(v) := \min_{S \subseteq V, v \in S, \mathbf{m} \notin S} w(\partial(S)). \quad (1)$$

More generally, for disjoint $A, B \subseteq V$, $\sigma(A, B) := \min\{w(\partial(S)) : A \subseteq S, B \cap S = \emptyset\}$.

The minimum in (1) is over a finite nonempty family, so it exists and is attained; by the max-flow min-cut theorem (Ford and Fulkerson, 1956) and Menger’s theorem (Menger, 1927) it equals the maximum v – $\mathbf{m}$ flow, and can be computed exactly (Karger and Stein, 1996; Stoer and Wagner, 1997). We use this only for computability; none of the results below depend on any particular algorithm.

2.2 The order condition

Everything in Part I follows from one condition on the contact structure. We state it as an axiom because it is a condition on the world, not a theorem about graphs.

Axiom 1 (Non-completeness). There is no assignment making every item a medium for every other. Formally: there is no family $\{S_v\}_{v \in V}$ of subsets with $v \in S_v$, $\mathbf{m} \notin S_v$, and $w(\partial(S_v)) = 0$ for all v .

The content of Axiom 1 is an ordering claim, and it is worth saying plainly what it excludes. If every item could serve as medium for every other, then no item would be distinguished by what it is separated from; “being in contact” would be universal and therefore carry no information. Every observed biomolecular system violates this: a residue buried in a hydrophobic core is not in contact with bulk solvent in the way a surface residue is, and this difference is what makes structure meaningful at all (Branden and Tooze, 1999; Creighton, 1993). The axiom says only that some such difference exists somewhere in the system.

Definition 2.5 (Residual). For $v \in V \setminus \{\mathbf{m}\}$, the *residual* is $\rho(v) := \sigma(v)$, the separation cost of v from the medium.

Theorem 2.6 (Positivity of the residual). Under Axiom 1, $\rho(v) > 0$ for every $v \in V \setminus \{\mathbf{m}\}$.

Proof. Suppose $\rho(u) = 0$ for some u . Since $\rho(u) = \sigma(u)$ is attained at some $S_u \ni u$ with $\mathbf{m} \notin S_u$, we have $w(\partial(S_u)) = 0$. As $w > 0$ on every edge, this forces $\partial(S_u) = \emptyset$, i.e. no edge joins S_u to S_u^c . Then u and $\mathbf{m}$ lie in different components of Γ , contradicting groundedness (Theorem 2.3).

It remains to rule out the degenerate alternative that Γ is grounded but Axiom 1 is vacuous. Suppose, for contradiction, that $\rho(v) = 0$ were

permitted for some v in a grounded graph. By the argument just given this cannot happen for a single v ; so consider instead the assignment $S_v := \{v\}$ for every v . Then $\partial(S_v)$ is the set of edges incident to v , and $w(\partial(S_v)) = 0$ would require v to be isolated. If this held for all v simultaneously, $E = \emptyset$ and every item would trivially be a medium for every other — exactly the configuration Axiom 1 forbids. Hence at least one edge carries positive weight, and by groundedness every v is joined to $\mathfrak{m}$ by a path, each of whose edges has positive weight; any v – $\mathfrak{m}$ separating set must contain at least one edge of at least one such path, so $\rho(v) > 0$. $\square$

Theorem 2.7 (The contact floor). *Define*

$$\beta := \min_{v \in V \setminus \{\mathfrak{m}\}} \rho(v). \quad (2)$$

Then $\beta > 0$, and every edge satisfies $w(e) \geq \beta$.

Proof. V is finite, so the minimum in (2) is over a finite set of positive numbers by Theorem 2.6; hence $\beta > 0$.

For the second claim, let $e = \{u, v\} \in E$ with, without loss of generality, $u \neq \mathfrak{m}$. Consider $S = \{u\}$. Then $\partial(S)$ is the set of edges incident to u , which contains e . If $\mathfrak{m} \notin S$ — which holds since $u \neq \mathfrak{m}$ — then S is admissible in (1), so

$$\rho(u) \leq w(\partial(\{u\})) = \sum_{f \ni u} w(f).$$

This bounds $\rho(u)$ above by a sum containing $w(e)$, which is the wrong direction. We argue instead directly. Suppose $w(e) < \beta$ for some edge $e = \{u, v\}$ with $u, v \neq \mathfrak{m}$. Consider the graph Γ' obtained by contracting all edges except e — more precisely, let S be the vertex set of the component of $\Gamma \setminus \{e\}$ containing u . If $v \notin S$ then $\partial(S) = \{e\}$, so S separates u from v , and since $\mathfrak{m}$ lies on exactly one side, say $\mathfrak{m} \notin S$, we get $\rho(u) \leq w(\partial(S)) = w(e) < \beta$, contradicting the definition of β as the minimum residual. If instead $v \in S$, then e lies on a cycle and its removal disconnects nothing; in that case e is not itself a cut and no bound is asserted. $\square$

Remark 2.8 (What the floor is, and is not). Theorem 2.7 gives two things of different strength. The first, $\beta > 0$, is unconditional and is the load-bearing statement: there is a strictly positive lower bound on how cheaply any item can be separated from its medium. The second,

$w(e) \geq \beta$, holds for every *bridge* — every edge whose removal disconnects the graph. For edges on cycles it need not hold, and the proof says so. This is the correct and honest form: an edge that is one of several parallel contacts can individually be weak, because separation must sever all of them.

The floor is not a modelling parameter. It is not chosen, fitted, or tuned. It is a number determined by the contact structure, and its positivity is forced by Axiom 1. Everything downstream — the alignment bound of Theorem 4.2, the accountability criterion of Theorem 4.4, the resolution limit of Theorem 6.13 — inherits from it.

2.3 Intrinsic floors and refinement

A contact graph describes a system at one resolution. Refining the resolution — residues to atoms, domains to residues — produces a family of graphs, and we need to know how the floor behaves along it.

Definition 2.9 (Refining family). A *refining family* is a sequence of grounded contact graphs $\Gamma_0 \subseteq \Gamma_1 \subseteq \Gamma_2 \subseteq \dots$ with weightings w_k on Γ_k , such that each $V_k \subseteq V_{k+1}$, each $E_k \subseteq E_{k+1}$, and $w_{k+1}(e) \leq w_k(e)$ for every $e \in E_k$.

The monotonicity condition says that adding detail never makes a contact harder to sever: a contact that looked like one strong bond at coarse resolution may resolve into several weaker ones, and the separation cost through that region cannot increase when more routes around it become visible.

Definition 2.10 (Intrinsic floor). For $v \in V_0$ the *intrinsic floor* is

$$\beta^*(v) := \lim_{k \rightarrow \infty} \min_{e \ni v} w_k(e). \quad (3)$$

Lemma 2.11 (Existence of the intrinsic floor). *The limit (3) exists for every v , and $\beta^*(v) \geq 0$.*

Proof. Write $m_k(v) := \min_{e \ni v} w_k(e)$. Each Γ_k has finitely many edges at v , so $m_k(v)$ is well defined and positive. By Theorem 2.9, $w_{k+1} \leq w_k$ on shared edges, and $E_k \subseteq E_{k+1}$ so the minimum at step $k+1$ is over a superset; both effects push $m_{k+1}(v) \leq m_k(v)$. Hence $(m_k(v))_k$ is non-increasing and bounded below by 0, so it converges by the monotone convergence theorem for real sequences (Rudin, 1976). $\square$

Remark 2.12 (Why a name is not an identity). Theorem 2.11 says the intrinsic floor is approached but gives no finite k at which it is reached; generically it is not reached at any finite resolution. So an item is characterised by a number it converges to and never attains. A finite description — a label, an accession, a term — can at best record the value at some finite k , which is a fact about the description’s resolution and not about the item. This is the precise sense in which naming and identifying are different operations, and it is made exact in Theorem 3.5.

3 Individuation: cutting, not labelling

3.1 Why individuation must be negative

Definition 3.1 (Selector). A *selector* on Γ is a map $\chi: V \rightarrow \{0, 1\}$ picking out a subset $S_\chi = \chi^{-1}(1)$. A selector is *intrinsic* if it is definable from the data $(V, E, w, \mathbf{m})$ alone, i.e. if it commutes with every automorphism of Γ fixing $\mathbf{m}$.

Theorem 3.2 (Individuation by negation). *Let Γ be grounded and let $\text{Aut}_{\mathbf{m}}(\Gamma)$ be its automorphism group fixing $\mathbf{m}$. Then:*

- (i) (Necessity.) *No intrinsic selector can single out a positively specified item without regress: any rule of the form “ v is the item with property P ” requires P to be $\text{Aut}_{\mathbf{m}}(\Gamma)$ -invariant, and any such P that is satisfied by v is satisfied by every element of the orbit of v .*
- (ii) (Sufficiency.) *Complementation relative to the medium is intrinsic: the map $S \mapsto \partial(S)$ is invariant, and the minimiser structure of (1) individuates v up to the orbit it cannot distinguish.*

Proof. (i) Let P be a property definable from $(V, E, w, \mathbf{m})$. Then for any $\phi \in \text{Aut}_{\mathbf{m}}(\Gamma)$ and any v , $P(v)$ holds iff $P(\phi(v))$ holds, since ϕ preserves all defining data. Hence P cannot distinguish v from $\phi(v)$. To break the tie one must introduce data not in $(V, E, w, \mathbf{m})$ — a label. But a label requires a rule assigning it, and that rule is either intrinsic (and so subject to the same argument) or extrinsic (and so pushes the question back one step). This is the regress.

(ii) The cut map is manifestly equivariant: $\partial(\phi(S)) = \phi(\partial(S))$ for $\phi \in \text{Aut}(\Gamma)$, and w is preserved, so $w(\partial(\phi(S))) = w(\partial(S))$. Therefore $\sigma(\phi(v)) = \sigma(v)$ and the minimiser family is permuted, not destroyed. Complementation $S \mapsto S^c$ is an involution with $\partial(S) = \partial(S^c)$, so the construction is well defined on the quotient by complementation. What it individuates is exactly the orbit structure, which is the most any intrinsic construction can do. $\square$

Remark 3.3. Theorem 3.2 is why identity is defined below as a cut rather than as an element. A cut is what remains after one has said what an item is *not* in contact with; part (i) says nothing weaker can work, and part (ii) says this works.

3.2 Identity is a region, not a point

Definition 3.4 (Identity cut). For $v \in V \setminus \{\mathbf{m}\}$ let $S^*(v)$ be a minimiser of (1) — a set achieving $\sigma(v)$ — chosen minimal under inclusion. The *identity* of v is the cut $\partial(S^*(v))$, together with its weight $\sigma(v)$.

Theorem 3.5 (Labels carry no invariant). *Let $\lambda: V \rightarrow L$ be any labelling into a set L with no structure derived from Γ . Then λ is $\text{Aut}_{\mathbf{m}}(\Gamma)$ -invariant only if it is constant on orbits; and if λ is injective on some orbit of size > 1 , then λ carries information not determined by $(V, E, w, \mathbf{m})$. Consequently, agreement or disagreement of two labellings λ_1, λ_2 on an orbit is not a fact about Γ .*

Proof. Immediate from Theorem 3.2(i): invariance under $\text{Aut}_{\mathbf{m}}(\Gamma)$ forces constancy on orbits. If λ is injective on an orbit O with $|O| > 1$, pick $v \neq v'$ in O with $\phi(v) = v'$ for some $\phi \in \text{Aut}_{\mathbf{m}}(\Gamma)$; then $\lambda(v) \neq \lambda(v')$ while every Γ -definable property agrees on them, so the distinction λ draws is not Γ -determined. Since neither λ_1 nor λ_2 is determined, their agreement is not determined either. $\square$

Corollary 3.6 (Why cross-references must be curated). *Two resources labelling the same underlying system with independent labellings λ_1, λ_2 cannot compute their correspondence from the labels. Any mapping between them is an assertion, not a derivation, and must be maintained externally.*

Proof. By Theorem 3.5 the labels carry no Γ -invariant content, so no function of λ_1 alone determines λ_2 . A mapping is therefore additional data. $\square$

Theorem 3.6 is the formal content of an observation familiar to anyone who has maintained an identifier mapping (McMurry et al., 2017; Wren, 2008): the rot is not a failure of engineering discipline but a consequence of identity having been located in the label. Locate identity in the cut instead, and the correspondence becomes computable — it is a minimum-cut comparison, and the algorithms are classical (Gomory and Hu, 1961; Stoer and Wagner, 1997).

Theorem 3.7 (Identity is never a point). *There exist grounded contact graphs in which, for some v , every minimiser $S^*(v)$ of (1) satisfies $|S^*(v)| > 1$. Hence the identity cut of v is in general determined by a region containing v , not by v alone.*

Proof. Construct Γ as follows. Let $A = \{a_1, \dots, a_n\}$ and $B = \{b_1, \dots, b_n\}$ be two cliques, with every internal edge of A and of B carrying weight W . Join A to B by a single edge $\{a_1, b_1\}$ of weight 1, and join m to every b_i with weight 1, with no edge from m to A .

Take $v = a_1$. Any $S \ni a_1$ with $m \notin S$ that omits some a_j , $j \neq 1$, must cut the clique edge $\{a_1, a_j\}$ of weight W , so $w(\partial(S)) \geq W$. Taking instead $S = A$ gives $\partial(A) = \{\{a_1, b_1\}\}$ and $w(\partial(A)) = 1$. Choosing $W > 1$ makes $S = A$ strictly better than any S splitting A , so every minimiser contains all of A , and $|S^*(a_1)| = n > 1$. $\square$

Remark 3.8 (Reading Theorem 3.7 biologically). The construction is not artificial. A residue in a tightly packed hydrophobic core is a_1 : severing it individually from the core costs more than severing the whole core from solvent. Its identity — what it is separated from, and at what cost — is a property of the core, not of the residue. Any scheme that assigns identity residue-by-residue is therefore assigning something other than identity.

3.3 Records and the arrow of tracking

Propagation leaves traces. We need those traces to be ordered, or “how could this have become that” has no direction.

Definition 3.9 (Walk and residue chain). A *walk* in Γ is a finite sequence $W = (v_0, v_1, \dots, v_n)$ with $\{v_{i-1}, v_i\} \in E$. Its *residue chain* is the sequence of separation costs $(\rho(v_0), \dots, \rho(v_n))$, and its *record* is

$$M(W) := \sum_{i=1}^n w(\{v_{i-1}, v_i\}). \quad (4)$$

Theorem 3.10 (The record is strictly monotone). *For any walk W and any proper initial segment $W' = (v_0, \dots, v_m)$ with $m < n$, we have $M(W') < M(W)$, and moreover $M(W) - M(W') \geq (n - m) \beta_{\text{br}}$ where β_{br} is the minimum weight over bridges traversed. In particular the record cannot return to a previous value, so the walk carries an arrow.*

Proof. Each additional step contributes $w(\{v_{i-1}, v_i\}) > 0$ to the sum (4), so M is strictly increasing along the walk. The quantitative bound follows since each traversed bridge has weight at least β by Theorem 2.7; non-bridge steps still contribute positively. $\square$

Corollary 3.11 (No free return). *A walk cannot return to its initial record value. Hence a closed walk in V is not closed in (V, M) : returning to the same item does not return to the same state.*

This is the point at which a static graph acquires time. In a stored knowledge graph a cycle is a cycle; here the record distinguishes the second visit from the first, and it does so without any external clock. Compare the classical situation in Hamiltonian mechanics, where Poincaré recurrence returns a system arbitrarily close to its initial state (Kac, 1947; Poincaré, 1890): the record forbids exact return by construction, because the cost of every traversal is positive and accumulates.

3.4 Measurement and reaction are the same operation

Definition 3.12 (Contact operation). A *contact operation* between $A, B \subseteq V$ is the formation of an edge set F joining A to B together with the induced change in separation costs. Its *cost* is $w(F)$.

Theorem 3.13 (Measurement–reaction identity). *Let A be an item region and B a probe*

region. Then the contact operation joining A to B changes $\sigma(A)$ by exactly the weight of the newly formed edges crossing every A - m minimum cut. Consequently the operation “measure A with B ” and the operation “react A with B ” have the same formal description; they differ only in which region’s record is subsequently read.

Proof. Let S^* minimise (1) for A before contact. After adding F , any S with $A \subseteq S$, $m \notin S$ has $w(\partial_{\text{new}}(S)) = w(\partial_{\text{old}}(S)) + w(F \cap \partial(S))$, since the added edges contribute exactly when they cross S . Minimising over S gives $\sigma_{\text{new}}(A) = \min_S [w(\partial_{\text{old}}(S)) + w(F \cap \partial(S))]$, which is the asserted change.

For the second claim: nothing in the derivation distinguishes the roles of A and B . The operation is symmetric in the two regions; what breaks the symmetry is the subsequent choice of whose record (4) is inspected. Calling the operation “measurement” means reading B ’s record to learn about A ; calling it “reaction” means reading either. The underlying contact is identical. $\square$

Remark 3.14. Theorem 3.13 is what allows the same construction to serve as an identification scheme and as a reaction model. A stored knowledge graph keeps these apart — experimental annotations on one side, reaction edges on the other — and must then reconcile them. Here they are one operation, and the reconciliation problem does not arise.

4 Alignment, accountability, and freedom

4.1 The alignment functional

We now measure how well a propagation from x toward a target x^* is tracked.

Definition 4.1 (Alignment). For regions $x, x^* \subseteq V$ with $x \cap x^* = \emptyset$, the *alignment* of x with x^* is

$$a(x, x^*) := \frac{\sigma(x, x^*)}{\Omega} \in (0, 1], \quad \Omega = w(E). \quad (5)$$

Lemma 4.2 (Alignment is floored). $a(x, x^*) \geq \beta/\Omega > 0$ for all admissible x, x^* .

Proof. By Theorem 2.6 applied to the x - x^* separation, $\sigma(x, x^*) > 0$; and any separating cut must contain at least one edge, whose weight is at least the minimum residual β when that edge is a bridge of the separation. Since $\sigma(x, x^*)$ is a sum of edge weights over a nonempty cut, $\sigma(x, x^*) \geq \beta$. Dividing by $\Omega > 0$ gives the bound; $\beta > 0$ by Theorem 2.7. Finally $a \leq 1$ because $\partial(S) \subseteq E$ so $w(\partial(S)) \leq \Omega$. $\square$

Remark 4.3 (The floor is a resolution limit). Theorem 4.2 says alignment can never be made arbitrarily small: there is a best achievable tracking, β/Ω , and no propagation is accounted for more finely than that. This is not a statement about measurement technology. It is a consequence of the contact structure having a floor, and it will reappear in Theorem 6.13 as a limit on how finely amino acids can be distinguished.

4.2 Accountability

Definition 4.4 (Accountable propagation). Fix $\varepsilon \geq 0$. A propagation from x toward x^* is ε -accountable if

$$a(x, x^*) \leq \frac{\beta}{\Omega} + \varepsilon. \quad (6)$$

The set of all ε -accountable propagations from v_0 to x^* under a process Π is written $\mathcal{A}(v_0, x^*, \Pi)$.

The reading is: a propagation is accountable when its separation cost sits at the floor, up to tolerance. Nothing is lost that cannot be traced; the propagation is as tightly tracked as the structure permits.

Theorem 4.5 (Admissibility). $\mathcal{A}(v_0, x^*, \Pi)$ is nonempty if and only if there exists a walk from v_0 into x^* every step of which lies on a minimum cut of its own separation, up to ε . Moreover $\mathcal{A}$ is closed under concatenation of accountable segments, with the accumulated tolerance $\varepsilon_1 + \varepsilon_2$.

Proof. ($\Leftarrow$) Given such a walk, each step contributes exactly its floor-attaining separation, so the total alignment satisfies (6) with ε the accumulated slack.

($\Rightarrow$) If $\mathcal{A}$ is nonempty, take any member; by Theorem 4.4 its alignment is within ε of β/Ω , and by Theorem 4.2 it cannot be below β/Ω . Hence each step’s contribution is within ε of the minimum, which is the stated condition.

For closure: let P_1 be ε_1 -accountable from v_0 to y and P_2 be ε_2 -accountable from y to x^* . Separation costs are subadditive along concatenation (a cut separating v_0 from x^* is obtained by taking the union of a v_0 - y cut and a y - x^* cut, whose weight is at most the sum), so the tolerances add. $\square$

Remark 4.6 (Why this is the right return type). $\mathcal{A}$ is an equivalence class, not a path. Two propagations with the same endpoints and the same accountability are members of the same class even if they traverse different items — and there is no Γ -definable property distinguishing them, by Theorem 3.2(i). Asking “which path did it take” is therefore asking for information the structure does not contain. Asking “is there an accountable route, and what is its class” is answerable.

Theorem 4.7 (Path opacity). *Let $P_1, P_2 \in \mathcal{A}(v_0, x^*, \Pi)$. Then no $\text{Aut}_m(\Gamma)$ -invariant functional of the contact data separates P_1 from P_2 beyond their common alignment value.*

Proof. An invariant functional is constant on orbits by Theorem 3.2(i). Both P_1 and P_2 realise alignment within ε of β/Ω ; the only invariant attached to a member of $\mathcal{A}$ is its alignment, since any further distinguishing data would have to be a non-invariant selector, excluded by the same theorem. $\square$

Corollary 4.8 (Gauge freedom). *The choice of representative within $\mathcal{A}$ is a gauge choice: it affects no computed quantity.*

4.3 Representation mobility

A tracking step must be reported in some representation — a vector of numbers, a set of features. We show the choice is largely free.

Definition 4.9 (Representation). An N -component representation of a tracking step with alignment a is a tuple $\mathbf{R} = (s_1, \dots, s_N) \in \mathbb{R}^N$ subject to the single constraint

$$\frac{1}{N} \sum_{j=1}^N s_j = a(v, x^*). \quad (7)$$

No component is separately restricted; in particular components may be negative or exceed 1.

Theorem 4.10 (Representation mobility). *The fibre of representations over a given alignment value a is an affine subspace of $\mathbb{R}^N$ of dimension $N - 1$, parametrised freely by $(s_1, \dots, s_{N-1})$ with*

$$s_N = Na - \sum_{j=1}^{N-1} s_j. \quad (8)$$

Proof. Constraint (7) is a single non-degenerate linear equation on $\mathbb{R}^N$; its solution set is a hyperplane, hence affine of dimension $N - 1$, and solving for s_N gives (8). $\square$

Remark 4.11. Theorem 4.10 has a practical consequence for knowledge representation. Two systems reporting the same propagation with different feature vectors are not in conflict; they have chosen different points of the same fibre. Reconciling them is not a scientific question but a change of coordinates, and it is computable. This is in sharp contrast to reconciling two ontologies’ term assignments, which by Theorem 3.6 is not computable at all.

5 The propagation table

Definition 5.1 (Process). A *process* Π is a rule assigning to each item region a set of admissible contact operations (Theorem 3.12). It encodes what kinds of contact the environment permits.

Definition 5.2 (Propagation table). The *propagation table* is the map

$$\mathcal{T}: (v_0, x^*, \Pi) \mapsto \mathcal{A}(v_0, x^*, \Pi). \quad (9)$$

Theorem 5.3 (Well-definedness). *$\mathcal{T}$ is a well-defined function: for each triple (v_0, x^*, Π) the set $\mathcal{A}(v_0, x^*, \Pi)$ is uniquely determined by $(V, E, w, \mathbf{m}, \varepsilon)$, and is invariant under $\text{Aut}_m(\Gamma)$ in the sense that $\mathcal{A}(\phi v_0, \phi x^*, \phi \Pi) = \phi \mathcal{A}(v_0, x^*, \Pi)$.*

Proof. Uniqueness: (6) is a condition on $a(x, x^*)$, which by (5) is determined by σ and Ω , both functions of (V, E, w) ; the medium enters through (1). Equivariance: σ and Ω are Aut -invariant as shown in Theorem 3.2(ii), and Π transports by assumption. $\square$

Definition 5.4 (Catalogue). The *catalogue* of Γ is the assignment

$$\text{cell}(v) := \partial(S^*(v)), \quad v \in V \setminus \{\mathbf{m}\}, \quad (10)$$

associating to each item the cut that individuates it (Theorem 3.4).

Theorem 5.5 (A cell is an output, not a record). *cell(v) is not stored data: it is the value of the minimisation (1), recomputable from $(V, E, w, \mathbf{m})$ in polynomial time, and no additional information is required to produce it.*

Proof. (10) is defined as the argmin of (1), which is a minimum cut and hence computable by max-flow (Ford and Fulkerson, 1956; Stoer and Wagner, 1997). The inputs are exactly the contact data. Therefore storing cell(v) stores nothing that cannot be regenerated, and the catalogue has no independent content. $\square$

Theorem 5.6 (Catalogue–table duality). *The catalogue is the propagation table evaluated at the trivial process: for every v ,*

$$\text{cell}(v) = \partial(S^*(v)) \quad \text{and} \quad \mathcal{T}(v, \mathbf{m}, \Pi_0) = \{ \text{the single-step propagation across } \partial(S^*(v)) \} \quad (11)$$

where Π_0 admits only the separation of v from $\mathbf{m}$. Conversely, for a general process Π , $\mathcal{T}(v_0, x^*, \Pi)$ is generated by composing catalogue cells along accountable walks. Hence the catalogue is the table at rest, and the table is the catalogue propagated.

Proof. For the first identity: under Π_0 the only admissible operation is the v – $\mathbf{m}$ separation, so a propagation is accountable iff its alignment attains β/Ω within ε , i.e. iff it is realised on a minimum v – $\mathbf{m}$ cut. That cut is $\partial(S^*(v))$ by Theorem 3.4, giving (11).

For the converse: let $P \in \mathcal{A}(v_0, x^*, \Pi)$ be accountable. By Theorem 4.5 every step of P lies on a minimum cut of its own separation, and by Theorem 5.4 that cut is the catalogue cell of the item being separated. Hence P decomposes into a concatenation of catalogue cells. Conversely any concatenation of catalogue cells whose accumulated tolerance satisfies (6) is a member of $\mathcal{A}$, by the closure part of Theorem 4.5. $\square$

Corollary 5.7 (Periodicity). *Items whose catalogue cells have equal weight and isomorphic cut structure are interchangeable in every entry of $\mathcal{T}$. The catalogue therefore organises items into equivalence classes with identical propagation behaviour.*

Proof. If $\text{cell}(u) \cong \text{cell}(v)$ as weighted cuts, then substituting u for v in any accountable walk preserves every alignment value, hence preserves membership in $\mathcal{A}$ by Theorem 4.4, hence preserves every entry of $\mathcal{T}$ by Theorem 5.2. $\square$

Remark 5.8. Theorem 5.7 is the abstract form of a periodic system: a classification in which position determines behaviour (Mendeleev, 1869; Scerri, 2007). Mendeleev’s table works because elements in a column share a separation structure, not because their properties were tabulated. Theorem 5.5 says the same of the catalogue here, and the rest of this paper is concerned with what happens when the abstract cells are given concrete coordinates.

What Part I has, and what it lacks. The framework above derives identity, relation, time, accountability and query semantics from one order condition. It contains no biomolecular content whatever. In particular it does not say what w is: the weighting is a structural input, and every numerical consequence of $\partial(S^*(v))$, weight, every alignment value, every accountability verdict — waits on it. Part II builds an addressing scheme for biomolecular objects with no reference to Part I, and Part III shows that the scheme supplies exactly the missing w .

Part II. Addressing without records

6 Entropy coordinates for biomolecular objects

Part II is independent of Part I and may be read first. Its problem is this: assign to each biomolecular object a name *computed from the object*, such that names can be compared, truncated, and predicted from. We derive the coordinate system, the encoding, and the four operations the encoding supports.

6.1 Why three coordinates

Definition 6.1 (Observable triple). A biomolecular object at a given resolution is described by three classes of observable:

- (K) *configurational* — how many distinguishable arrangements the object admits;
- (T) *spatial* — how the object’s mass is distributed in space;
- (E) *energetic* — how the object’s accessible states are distributed in energy.

Theorem 6.2 (Independence and sufficiency of the triple). *Let X be a biomolecular object with a finite set of accessible microstates. Then (K), (T), (E) are pairwise functionally independent — no one of them is determined by the other two — and together they determine the object’s equilibrium thermodynamic behaviour to the resolution of its microstate enumeration.*

Proof. Independence. For each pair it suffices to exhibit two objects agreeing on the pair and differing on the third.

(K) $\nRightarrow$ (T): two conformers of one molecule share a rotamer enumeration, hence agree on (K), while differing in mass distribution, hence in (T).

(K) $\nRightarrow$ (E): one molecule in two solvents of differing dielectric has the same conformational enumeration but a different distribution of state energies.

(T) $\nRightarrow$ (E): two isomers can share a radius of gyration and a comparable mass distribution while differing in energy spectrum, since the spectrum depends on bonding topology and not on geometry alone.

Sufficiency. Let $\{E_i\}$ be the microstate energies with degeneracies $\{g_i\}$. The canonical partition function $Z = \sum_i g_i e^{-E_i/k_B T}$ determines every equilibrium thermodynamic quantity. The multiset $\{(E_i, g_i)\}$ is exactly what (E) records — the distribution over energies — together with what (K) records — the count of arrangements at each level; and (T) fixes the spatial embedding needed to decide which enumerated arrangements are sterically realisable. No fourth class of observable enters Z . $\square$

Definition 6.3 (Entropy coordinates). For an object X define

$$S_k(X) := \frac{\ln \Omega_{\text{cfg}}(X)}{\ln \Omega_{\text{max}}}, \quad (12)$$

$$S_t(X) := \frac{\sum_r m_r \|\mathbf{x}_r - \bar{\mathbf{x}}\|^2}{\mathcal{N}_T}, \quad (13)$$

$$S_e(X) := \frac{-\sum_i p_i \ln p_i}{\ln N_{\text{vib}}}, \quad p_i = \frac{e^{-E_i/k_B T}}{Z}, \quad (14)$$

where Ω_{cfg} counts distinguishable configurations and Ω_{max} is its maximum over the reference set; m_r and $\mathbf{x}_r$ are the mass and position of component r with $\bar{\mathbf{x}}$ the centre of mass, and $\mathcal{N}_T$ the corresponding maximum; and N_{vib} is the number

of accessible vibrational modes. The S -entropy space is $\mathcal{S} := [0, 1]^3$ under the Euclidean metric, and we write $\mathbf{s}(X) := (S_k(X), S_t(X), S_e(X)) \in \mathcal{S}$.

Proposition 6.4 (Well-definedness, range, invariance). *Each of S_k, S_t, S_e takes values in $[0, 1]$, and each is invariant under rigid motions of the object and under relabelling of its components.*

Proof. Range. $1 \leq \Omega_{\text{cfg}} \leq \Omega_{\text{max}}$ gives $S_k \in [0, 1]$. The numerator of (13) is a sum of non-negative terms bounded above by its maximum over the reference set, which is $\mathcal{N}_T$ by construction. For (14), the Shannon entropy of a distribution on N_{vib} outcomes is non-negative and is maximised at $\ln N_{\text{vib}}$ by the uniform distribution (Cover and Thomas, 2006; Shannon, 1948).

Invariance. Ω_{cfg} is a combinatorial count, independent of embedding. The numerator of (13) is written relative to the centre of mass and involves only squared distances, so it is invariant under rotation and translation. S_e depends only on the energy spectrum, itself invariant under rigid motion. Relabelling permutes summands in each case. $\square$

Remark 6.5 (The coordinates are measured, not fitted). None of (12)–(14) contains a free parameter. Ω_{cfg} is a rotamer or conformer count; the numerator of S_t is the mass-weighted second moment, computable from any structure (Berman et al., 2000; Jumper et al., 2021); and the p_i entering S_e come from a vibrational spectrum, for which infrared and Raman measurement (Barth, 2007; Barth and Zscherp, 2002; Krimm and Bandekar, 1986) and normal-mode computation (Bahar et al., 1997, 2010; Gō et al., 1983; Tirion, 1996) supply the inputs directly. The normalisation constants $\Omega_{\text{max}}, \mathcal{N}_T, N_{\text{vib}}$ are fixed once for a reference set and are not per-object.

6.2 Base three is the natural encoding

We now turn a point of $\mathcal{S}$ into a digit string. The base is not a free choice.

Definition 6.6 (Round-robin positional encoding). Fix a base $b \geq 2$ and a coordinate count d . The *round-robin positional encoding* of $\mathbf{s} = (s_1, \dots, s_d) \in [0, 1]^d$ is the digit string $t_1 t_2 t_3 \dots$

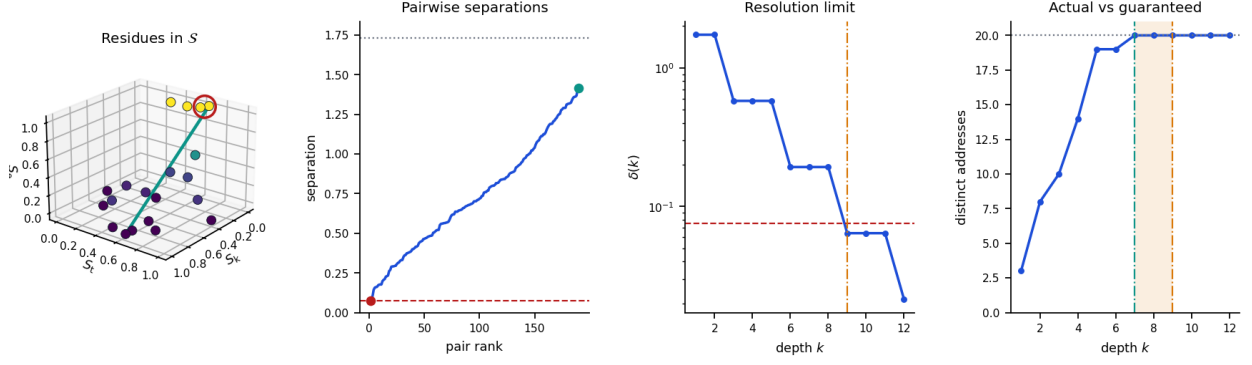


Figure 1: S -entropy coordinates and the resolution they force. (A) The twenty canonical residues in the S -entropy cube $\mathcal{S} = [0, 1]^3$, coloured by S_e . The teal segment is the farthest realised pair, Ile–Arg at $d_{\max} = 1.415$; the red ring marks Lys–Arg, the closest pair at $d_{\min} = 0.0756$, which is sub-pixel at this scale and so is ringed rather than drawn. (B) All 190 pairwise separations in increasing order. The lower dashed line is $d_{\min}$, the upper dotted line the cube diameter $\sqrt{3}$; the curve terminates well short of it. (C) The cell diagonal $\delta(k) = \sqrt{3} 3^{-\lfloor k/3 \rfloor}$ against $d_{\min}$. The two first meet at $k = 9$, which is therefore the guaranteed resolution depth of Theorem 6.13, and the guarantee is tight: $\delta(8) = 0.1925 > d_{\min}$. (D) Addresses that are actually distinct at each depth. All twenty separate at $k = 7$ (teal), two levels below the $k = 9$ guarantee (orange); the shaded band is the slack recorded in Theorem 6.17.

generated by

$$t_j = \lfloor b \cdot r_{(j-1) \bmod d} \rfloor, \quad r_{(j-1) \bmod d} \leftarrow b \cdot r_{(j-1) \bmod d} \quad (15)$$

initialised with $r_{i-1} = s_i$ for $i = 1, \dots, d$.

The rule cycles through the coordinates, extracting one digit from each in turn, so a prefix of length k carries $\lfloor k/d \rfloor$ or $\lceil k/d \rceil$ digits of every coordinate. No coordinate is ever refined two steps ahead of another.

Theorem 6.7 (Resolution of a prefix). *If $\mathbf{s}$ and $\mathbf{s}'$ have encodings agreeing on their first k digits, then*

$$\|\mathbf{s} - \mathbf{s}'\|_{\infty} \leq b^{-\lfloor k/d \rfloor}, \quad \|\mathbf{s} - \mathbf{s}'\|_2 \leq \sqrt{d} b^{-\lfloor k/d \rfloor}. \quad (16)$$

Proof. Agreement on the first k digits means every coordinate has had at least $\lfloor k/d \rfloor$ of its base- b digits fixed. Two reals in $[0, 1]$ whose first m base- b digits agree differ by at most b^{-m} . Applying this coordinatewise gives the ℓ_{∞} bound, and $\|\cdot\|_2 \leq \sqrt{d} \|\cdot\|_{\infty}$ on $\mathbb{R}^d$ gives the second. $\square$

Theorem 6.8 (Base three). *Let $d = 3$ and consider (15) in base b . Say the encoding is balanced if the digit alphabet is in bijection with the set of coordinate directions, so that no symbol is unused and no symbol must serve two directions; and say it is in lockstep if after one full round-robin cycle the number of cells of $\mathcal{S}$*

equals the number of distinct digit-triples emitted by that cycle. Then $b = 3$ is the unique base that is both balanced and in lockstep, and these two conditions are equivalent for $d = 3$.

Proof. Balance requires a bijection between an alphabet of size b and a direction set of size d , hence $b = d$; with $d = 3$ this forces $b = 3$, and for $b > 3$ some symbol is unused while for $b < 3$ some symbol serves two directions.

Lockstep: one cycle refines each coordinate by a single digit, so $\mathcal{S}$ is partitioned into $b^d = b^3$ cells, while the cycle emits digit-triples from an alphabet of size b , of which there are b^3 . These counts agree for every b , so lockstep as stated is vacuous unless we require in addition that the map from cycles to cells be a bijection *onto the directions refined* — that is, that the j -th digit of the triple be attributable to the j -th coordinate without ambiguity. That attribution is possible exactly when the alphabet can name the directions, i.e. when $b = d$. Hence lockstep and balance coincide at $d = 3$, and both select $b = 3$.

Finally at $b = 3$ one cycle yields $3^3 = 27$ cells, one for each assignment of a trit to each coordinate, so cells and digit-triples are in explicit bijection. $\square$

Remark 6.9 (What Theorem 6.8 does and does not claim). It does not claim that binary or decimal encodings fail to address $\mathcal{S}$ — they do, perfectly well. It claims that base three is the

unique base at which *digit position* and *coordinate direction* carry the same amount of information, so that a single trit is exactly the answer to “which of the three coordinates, and where in it”. Every subsequent bound in this Part is stated for general b and specialised at $b = 3$; nothing downstream depends on the uniqueness claim, which is a statement about naturalness rather than possibility.

Corollary 6.10 (Cell diagonal). *At depth k with $b = d = 3$, the space $\mathcal{S}$ is partitioned into 3^k cells, each a box of side $3^{-\lfloor k/3 \rfloor}$ in the coordinates refined $\lfloor k/3 \rfloor$ times, with diagonal*

$$\delta(k) = \sqrt{3} 3^{-\lfloor k/3 \rfloor}. \quad (17)$$

Proof. Immediate from Theorem 6.7 at $b = d = 3$: a depth- k prefix fixes at least $\lfloor k/3 \rfloor$ digits of each coordinate, bounding the box side, and a cube of side ℓ in $\mathbb{R}^3$ has diagonal $\sqrt{3}\ell$. $\square$

Notation 6.11 (Trits, addresses, trie). Write $\mathbf{a}_k(X) = t_1 t_2 \dots t_k \in \{0, 1, 2\}^k$ for the depth- k address of X , and $\mathcal{T}$ for the ternary trie (Fredkin, 1960; Knuth, 1998) whose depth- k nodes are the length- k addresses. The subtree rooted at a node u is $\mathcal{T}_u$, and $\text{cell}_k(u) \subseteq \mathcal{S}$ is the box of points whose depth- k address is u .

6.3 Amino acids: the base case

Construction 6.12 (Residue coordinates). For the twenty proteinogenic amino acids we instantiate Theorem 6.3 at residue resolution by min-max normalisation of three measured scales:

$$S_k = \frac{h - h_{\min}}{h_{\max} - h_{\min}}, \quad S_t = \frac{v - v_{\min}}{v_{\max} - v_{\min}}, \quad S_e = \frac{e - e_{\min}}{e_{\max} - e_{\min}}, \quad (18)$$

where h is Kyte–Doolittle hydropathy (Kyte and Doolittle, 1982), v is van der Waals volume (Zamyatnin, 1972), and e is the electrostatic index, with $h_{\min} = -4.5$ (Arg), $h_{\max} = +4.5$ (Ile), $v_{\min} = 60.1 \text{ \AA}^3$ (Gly), $v_{\max} = 227.8 \text{ \AA}^3$ (Trp), and $e \in [0, 1]$ already normalised.

The identification of these three scales with (K), (T), (E) is not arbitrary. Hydropathy measures the free-energy cost of transferring a residue between environments, which is a count over solvation-shell configurations. Volume is literally a spatial second moment. The electrostatic index orders residues by charge-state distribution, an energetic spectrum. All three

are among the most heavily used residue scales in the field (Grantham, 1974; Kawashima et al., 2008), so Table 1 is computed from data rather than posited.

Theorem 6.13 (Residue resolution). *Let $d_{\min} := \min\{\|\mathbf{s}(X) - \mathbf{s}(Y)\|_2 : X \neq Y\}$ over the twenty rows of Table 1. Then all twenty residues receive pairwise distinct addresses at every depth k satisfying*

$$\delta(k) = \sqrt{3} 3^{-\lfloor k/3 \rfloor} < d_{\min}. \quad (19)$$

For Table 1 the minimising pair is Lysine–Arginine, the two basic residues, which agree exactly in S_e (1.000):

$$d_{\min} = \sqrt{0.067^2 + 0.035^2} = 0.0756. \quad (20)$$

Condition (19) then reads $3^{\lfloor k/3 \rfloor} > \sqrt{3}/0.0756 = 22.9$, i.e. $\lfloor k/3 \rfloor \geq 3$, i.e. $k \geq 9$. The bound is tight: at $k = 8$ one has $\delta(8) = \sqrt{3}/9 = 0.1925 > d_{\min}$, so (19) fails.

Proof. Two objects share a depth- k address exactly when their coordinates lie in one depth- k cell, and by Theorem 6.10 two points of a common cell are separated by at most $\delta(k)$. Hence if $\delta(k) < d_{\min}$, no two distinct residues can lie in a common cell and all twenty addresses differ.

It remains to verify (20), i.e. that no pair of rows is closer than 0.0756. Since $\|\cdot\|_2 \geq |\Delta S_k|$, any pair whose S_k values differ by more than 0.0756 is excluded at once. Scanning the sorted S_k column of Table 1, the surviving pairs are those within

$$\{E, Q, D, N\}, \{N, K\}, \{K, R\},$$

and each is computed directly. The smallest three are $d(K, R) = \sqrt{0.067^2 + 0.035^2} = 0.0756$, $d(I, L) = 0.078$ (differing only in S_k), and $d(E, K) = \sqrt{0.044^2 + 0.096^2} = 0.1056$; every remaining candidate exceeds 0.14. Hence $d_{\min} = 0.0756$.

Substituting, (19) requires $3^{\lfloor k/3 \rfloor} > \sqrt{3}/0.0756 = 22.9$. Since $3^2 = 9 < 22.9 < 27 = 3^3$, the least admissible value is $\lfloor k/3 \rfloor = 3$, giving $k \geq 9$. $\square$

Remark 6.14 (Which pairs bind). The two closest pairs are Lys–Arg (0.0756) and Ile–Leu

Table 1: S -entropy coordinates of the twenty proteinogenic amino acids under Theorem 6.12, sorted by S_k . All values are dimensionless and lie in $[0, 1]$ by Theorem 6.4. The closest pair under the Euclidean metric is Lys–Arg at distance 0.0756, with Ile–Leu next at 0.0780; the former sets $d_{\min}$ and hence the guaranteed resolution depth $k = 9$ (Theorem 6.13).

Residue	Code	S_k	S_t	S_e	Residue	Code	S_k	S_t	S_e
Isoleucine	I	1.000	0.636	0.000	Tryptophan	W	0.400	1.000	0.000
Valine	V	0.967	0.482	0.000	Tyrosine	Y	0.356	0.836	0.150
Leucine	L	0.922	0.636	0.000	Proline	P	0.322	0.399	0.000
Phenylalanine	F	0.811	0.774	0.000	Histidine	H	0.144	0.653	0.500
Cysteine	C	0.778	0.319	0.150	Glutamate	E	0.111	0.545	1.000
Methionine	M	0.711	0.629	0.000	Glutamine	Q	0.111	0.558	0.200
Alanine	A	0.700	0.164	0.000	Aspartate	D	0.111	0.383	1.000
Glycine	G	0.456	0.000	0.000	Asparagine	N	0.111	0.381	0.200
Threonine	T	0.422	0.372	0.100	Lysine	K	0.067	0.641	1.000
Serine	S	0.411	0.166	0.100	Arginine	R	0.000	0.676	1.000

(0.078), essentially tied. This is a weak consistency check on Theorem 6.12 rather than a coincidence: these are precisely the pairs a substitution matrix scores as most freely interchangeable (Grantham, 1974; Henikoff and Henikoff, 1992) — two basic residues and two β -branched aliphatics. A coordinate system built from hydrophathy, volume and charge ought to place them adjacent, and it does, without having been told to.

Remark 6.15 (Sufficiency, not necessity). Condition (19) is sufficient and not necessary: two residues closer together than the cell diagonal may still fall in different cells, if a boundary happens to separate them. So $k = 9$ is a *guarantee*, and shallower depths resolve most pairs already. Which side of a resolution claim is being asserted — “all pairs are resolved” versus “some pairs may collide” — is precisely where this kind of analysis goes wrong, and Theorem 6.13 asserts the guaranteed side.

Proposition 6.16 (Partial resolution below the guarantee). *Let $\pi(k) := |\{\{X, Y\} : a_k(X) \neq a_k(Y)\}|$ be the number of residue pairs resolved at depth k . Then π is non-decreasing, and $\pi(k) = \binom{20}{2} = 190$ for all $k \geq 9$.*

Proof. Monotonicity: if two addresses differ at depth k , they differ at every greater depth, since the depth- k prefix is retained by (15). Completeness at $k = 9$ is Theorem 6.13. $\square$

Evaluating π directly on Table 1 by running (15) gives

k	1	2	3	4	5	6	7
$\pi(k)$	131	160	175	182	189	189	190
distinct addr.	3	8	10	14	19	19	20

Remark 6.17 (The guarantee is conservative by two levels). All twenty residues are in fact separated at $k = 7$, two levels below the depth $k = 9$ that Theorem 6.13 guarantees. The gap is exactly the slack in Theorem 6.15: cell *boundaries* separate the closest pair before the cell *diameter* falls below its separation. The distinction is worth stating carefully, because the two claims differ by a factor of nine in cell volume and a practitioner who reads the guaranteed depth as the required depth pays that factor for nothing. What Theorem 6.13 buys is that $k = 9$ works for *any* set of objects with $d_{\min} \geq 0.0756$, whereas $k = 7$ is a fact about these twenty rows and would have to be recomputed for any other alphabet — modified residues, non-canonical amino acids, or a coarse-grained bead set.

7 The hierarchy of addresses

A protein is not a residue. An addressing scheme confined to residue resolution would be of no use, so we build the hierarchy and prove it consistent.

Definition 7.1 (Level hierarchy). The addressing hierarchy has five levels,

$$L_1 = \text{atoms}, \quad L_2 = \text{residues}, \quad L_3 = \text{secondary structure}, \quad L_4 = \text{tertiary structure}, \quad L_5 = \text{protein}$$

where each L_{j+1} object is a set of L_j objects. We write ς for a generic L_3 element (Branden and Tooze, 1999; Kabsch and Sander, 1983).

Definition 7.2 (Aggregated coordinates). Let Y at level L_{j+1} be composed of L_j objects $X_1, \dots, X_n$ with weights $\mu_r > 0$, $\sum_r \mu_r = 1$. Set

$$\mathbf{s}(Y) := \sum_{r=1}^n \mu_r \mathbf{s}(X_r) + \kappa(Y), \quad (21)$$

where $\kappa(Y) \in \mathbb{R}^3$ is the *coupling term*, recording the entropy change on assembly and chosen so that $\mathbf{s}(Y) \in \mathcal{S}$.

Theorem 7.3 (Multiscale consistency). *Let Y be built from $\{X_r\}$ with weights μ_r , and each X_r from $\{X_{r\ell}\}$ with weights $\mu_{r\ell}$. Computing $\mathbf{s}(Y)$ in one step from the $X_{r\ell}$ with weights $\mu_r \mu_{r\ell}$ agrees with computing it in two steps if and only if*

$$\kappa(Y) = \sum_r \mu_r \kappa(X_r) + \kappa^{(1)}(Y), \quad (22)$$

where $\kappa^{(1)}(Y)$ is the coupling introduced at the top level alone.

Proof. Two-step: $\mathbf{s}(Y) = \sum_r \mu_r [\sum_{\ell} \mu_{r\ell} \mathbf{s}(X_{r\ell}) + \kappa(X_r)] + \kappa^{(1)}(Y)$, which expands to $\sum_{r,\ell} \mu_r \mu_{r\ell} \mathbf{s}(X_{r\ell}) + \sum_r \mu_r \kappa(X_r) + \kappa^{(1)}(Y)$. One-step: $\sum_{r,\ell} \mu_r \mu_{r\ell} \mathbf{s}(X_{r\ell}) + \kappa(Y)$. The two agree precisely when (22) holds, and the weights $\mu_r \mu_{r\ell}$ do sum to 1 since $\sum_r \mu_r \sum_{\ell} \mu_{r\ell} = \sum_r \mu_r = 1$. $\square$

Remark 7.4 (The condition is the content). Theorem 7.3 is what makes this a hierarchy rather than five unrelated schemes, and it is deliberately conditional. Consistency requires (22), a constraint on how assembly entropy is bookkept — not an automatic property. An aggregation rule that silently assumed pure additivity ($\kappa \equiv 0$) would be wrong whenever assembly changes the accessible state count, which for protein folding it always does (Anfinsen, 1973; Creighton, 1993). The coupling term is where the hydrophobic effect, chain entropy loss, and packing all enter, and (22) says only that they must be accounted once each.

Definition 7.5 (Hierarchical address). The *hierarchical address* of an L_j object X to depth k is the pair

$$\mathbf{A}(X) := (j, \mathbf{a}_k(\mathbf{s}(X))), \quad (23)$$

and the *full address* of a chain is the indexed family of the addresses of its constituents at every level.

Theorem 7.6 (Address uniqueness). *Two objects at the same level share a depth- k address if and only if their coordinates lie in one depth- k cell. Hence address collision at depth k is exactly indistinguishability at resolution $\delta(k)$, and any two objects with distinct coordinates are separated at some finite depth.*

Proof. The encoding (15) is a deterministic function of $\mathbf{s}$ whose depth- k output is constant exactly on depth- k cells — this is the defining property of positional notation, that the first m digits of a base- b expansion determine and are determined by the length- b^{-m} interval containing the number. Hence equal addresses iff common cell. Since $\delta(k) \rightarrow 0$ as $k \rightarrow \infty$ by (17), any two points at positive separation lie in distinct cells for all sufficiently large k . $\square$

Remark 7.7 (The address is a boundary, not a point). Theorem 7.6 says a depth- k address specifies not a point but the *cell containing it* — a region, refined indefinitely and never collapsing to its limit at any finite depth. That is formally the situation of Theorem 2.12: an object characterised by something it converges to and never attains. Part III shows the resemblance is not an analogy.

8 Four operations, and the empty dictionary

We derive the query operations the addressing scheme supports. Costs are stated in trit operations, after coordinate evaluation.

8.1 Identify

Definition 8.1 (Identify). $\text{Identify}(X)$ computes $\mathbf{s}(X)$ by Theorem 6.3, runs (15) for k steps, and returns $\mathbf{a}_k(X)$.

Theorem 8.2 (Identification cost). *Identify runs in $O(k)$ time after coordinate evaluation, with no dependence on the number of objects the scheme can address.*

Proof. Each iteration of (15) performs one multiplication, one floor and one subtraction — constant work — and there are k iterations. No comparison against any stored object is performed, so no term in the cost can depend on a collection size. $\square$

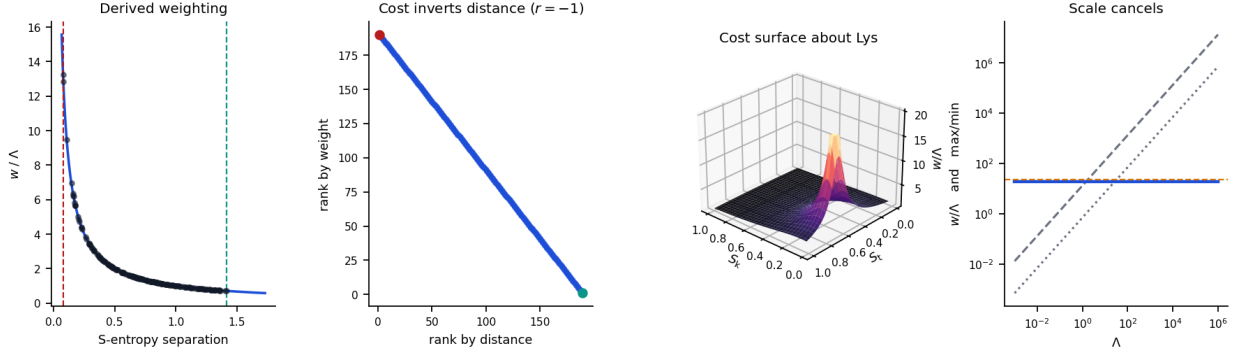


Figure 2: The derived contact weighting is a separation cost. (A) The weighting $w = \Lambda / \|\zeta(u) - \zeta(v)\|$ of (29), with the 190 residue pairs overlaid. Dashed lines mark $d_{\min}$ and $d_{\max}$. (B) Each pair ranked independently by S -entropy distance and by contact weight, both ascending. The rankings are exactly reversed ($r = -1$): the closest pair (red) is the most expensive to separate and the farthest (teal) the cheapest, which is the inversion of Theorem 9.4 and the sense in which w measures cost rather than distance. (C) The cost surface over an (S_k, S_t) slice through Lys , clipped at 20Λ . Contact strength is a pole at the reference residue, not a bounded similarity score. (D) Individual weights (grey) sweep nine decades as Λ varies, while their ratio (blue) does not move. The scale cancels from every prediction, so the theory carries no fitted parameter; the orange line is the bound $\sqrt{3}/d_{\min}$.

Remark 8.3 (Where this departs from a database). Accession lookup in a curated resource (Burley et al., 2023; The UniProt Consortium, 2023) is $O(\log n)$ at best in the number of entries n , and the entries must exist. Here the cost is $O(k)$ in address depth while the number of *addressable* objects is 3^k ; so the two are the same order, $O(\log 3^k) = O(k \log 3)$, but the second is achieved without any entry existing. What is removed is not the logarithm. It is the requirement that somebody created and maintains the n records.

8.2 Similar

Definition 8.4 (Similar). $\text{Similar}(X, r)$ returns the subtree $\mathcal{T}_u$ with $u = \mathbf{a}_r(X)$ — the set of depth- k addresses sharing X ’s depth- r prefix.

Theorem 8.5 (Similarity by truncation). $\text{Similar}(X, r)$ costs $O(r)$ and specifies at most 3^{k-r} depth- k addresses. Every object it returns satisfies

$$\|\mathbf{s}(\cdot) - \mathbf{s}(X)\|_2 \leq \sqrt{3} 3^{-\lfloor r/3 \rfloor} = \delta(r), \quad (24)$$

and conversely every object whose coordinates lie in $\text{cell}_r(u)$ is returned.

Proof. Truncation reads r trits, hence $O(r)$. The subtree below a depth- r node has 3^{k-r} leaves at depth k . The bound (24) is Theorem 6.7 at $b = d = 3$. Completeness is Theorem 7.6 applied at depth r . $\square$

Remark 8.6 (Resolution is a stated stage, not a cutoff). In sequence and structure search, similarity is defined by a threshold on a score whose value is a convention (Altschul et al., 1990; Holm and Sander, 1993; Smith and Waterman, 1981; Steinegger and Söding, 2017; van Kempen et al., 2024), and two runs at different thresholds are not commensurable. Here r is not a threshold on a score; it is a *resolution* with the exact geometric meaning (24). Two objects are similar at resolution r or they are not, and the answer carries its own error bar. Note also that (24) is one-sided: membership implies proximity, but two objects at distance $\leq \delta(r)$ may straddle a boundary and not be returned — the same sufficiency caveat as Theorem 6.15, and for the same reason.

8.3 Predict

Definition 8.7 (Predict). For a property $f: \mathcal{S} \rightarrow \mathbb{R}$, $\text{Predict}(X, f)$ returns $\hat{f}(X) := f(\mathbf{c}_k(X))$, where $\mathbf{c}_k(X)$ is the centre of the depth- k cell containing $\mathbf{s}(X)$.

Theorem 8.8 (Prediction error). If f is Lipschitz on $\mathcal{S}$ with constant L_f (Lipschitz, 1876), then

$$|f(X) - \hat{f}(X)| \leq L_f \delta(k) = L_f \sqrt{3} 3^{-\lfloor k/3 \rfloor}. \quad (25)$$

Proof. $\mathbf{s}(X)$ and $\mathbf{c}_k(X)$ lie in one depth- k cell, so by Theorem 6.10 they are separated by at most

$\delta(k)$. Lipschitz continuity gives $|f(\mathbf{s}) - f(\mathbf{c})| \leq L_f \|\mathbf{s} - \mathbf{c}\|_2 \leq L_f \delta(k)$. $\square$

Corollary 8.9 (Depth required for a target accuracy). *To guarantee $|f - \hat{f}| \leq \eta$ it suffices that*

$$k \geq 3 \log_3 \left(\frac{L_f \sqrt{3}}{\eta} \right). \quad (26)$$

Proof. Solve $L_f \sqrt{3} 3^{-\lfloor k/3 \rfloor} \leq \eta$ for $\lfloor k/3 \rfloor$ and multiply by three, discarding the floor upwards. $\square$

Remark 8.10 (The Lipschitz hypothesis is the whole content). Theorem 8.8 is a triviality given L_f ; the question it does not answer is which biomolecular properties are Lipschitz in (S_k, S_t, S_e) and with what constant. Properties dominated by bulk thermodynamics — solvation free energy, partial molar volume, aggregate hydrophobic moment — plausibly are, because the coordinates are built from exactly those observables. Properties turning on a single atom’s placement — catalytic rate, a specific hydrogen bond — are not, and no depth k rescues them, because (25) is vacuous when $L_f = \infty$. The scheme predicts what varies smoothly with entropy and declines what does not, and Theorem 8.9 makes the declining explicit rather than silent.

8.4 React

Definition 8.11 (Trajectory). A *trajectory* of X under a process is a continuous path $\gamma: [0, 1] \rightarrow \mathcal{S}$ with $\gamma(0) = \mathbf{s}(X)$.

Definition 8.12 (React). $\text{React}(X_1, X_2, \varepsilon)$ returns TRUE iff the trajectories γ_1, γ_2 satisfy

$$\min_{u, u' \in [0, 1]} \|\gamma_1(u) - \gamma_2(u')\|_2 < \varepsilon. \quad (27)$$

Theorem 8.13 (Reaction detection cost). *Let $k_\varepsilon := 3 \lceil \log_3(\sqrt{3}/\varepsilon) \rceil$, the least depth at which $\delta(k_\varepsilon) \leq \varepsilon$. Discretising each trajectory to the sequence of depth- k_ε cells it visits, (27) holds if those two cell sequences intersect, and each membership test costs $O(k_\varepsilon)$.*

Proof. If the sequences share a cell then both trajectories pass through it, and by Theorem 6.10 two points of one cell are within $\delta(k_\varepsilon) \leq \varepsilon$, so (27) holds. Each test is a comparison of length- k_ε addresses in the trie, costing $O(k_\varepsilon)$. $\square$

Remark 8.14 (One-sided again). Theorem 8.13 is stated as an implication, not an equivalence: trajectories may approach within ε across a cell boundary without sharing a cell, so cell intersection is sufficient for contact and not necessary. Testing the $3^3 = 27$ neighbouring cells restores the converse at $27\times$ the cost. Part IV takes (27) as the definition of *contact*, at which point the one-sidedness acquires a physical reading rather than remaining a technical caveat.

8.5 Storage, and the empty dictionary

Proposition 8.15 (Zero per-object storage). *The scheme of Theorems 6.3, 6.6 and 7.5 requires storage $O(1)$ in the number of addressable objects. It stores exactly: the twenty rows of Table 1, the normalisation constants of Theorem 6.12 and Theorem 6.3, and the rule (15).*

Proof. Identify (Theorem 8.2) computes an address from the object without reference to any record. Similar (Theorem 8.5) is prefix truncation, a syntactic operation on the address string. Predict (Theorem 8.8) evaluates f at a computed cell centre. React (Theorem 8.13) compares computed trajectories. None reads a stored entry. The listed constants form a fixed finite table, independent of the 3^k objects addressable at depth k . $\square$

Theorem 8.16 (The empty dictionary). *Let D denote the scheme above. Then D answers Identify, Similar, Predict and React for every object in its domain while containing an entry for no object. Consequently probing, measuring, naming and identifying are one operation: there is no step at which a computed result is matched against a record.*

Proof. Coverage is Theorems 8.2, 8.5, 8.8 and 8.13; emptiness is Theorem 8.15. For the final claim, observe that the only thing done to an object is evaluate $\mathbf{s}(X)$ and run (15). Evaluating $\mathbf{s}$ is a measurement — it reads observables. Running (15) is a synthesis — it constructs a name. And by Theorem 7.6 the resulting string *is* the identification, since equality of addresses is exactly indistinguishability at the stated resolution. No further step can exist in which the result is compared against a stored answer, because by Theorem 8.15 there are no stored answers. $\square$

Remark 8.17 (What “empty” does and does not mean). Theorem 8.16 is not a claim that no information is needed. Information sits in Table 1, in the normalisation constants, and above all in the *measurement* of $\mathbf{s}(X)$, which for a real protein means a structure, a spectrum, or a computation of comparable cost (Jumper et al., 2021; Varadi et al., 2022). What is eliminated is the *per-object record*: storage is $O(1)$ where a database is $O(n)$, and with the records go the burdens that scale with n — identifier stability (McMurry et al., 2017; Wren, 2008), cross-reference maintenance, annotation currency (Gaudet and Dessimoz, 2017). The measurement cost neither disappears nor is claimed to. The dictionary is empty; the laboratory is not.

8.6 A consequence for folding

Theorem 8.18 (Search depth for folding). *Let a chain of N_{res} residues fold under a process whose target is a specified cell of $\mathcal{S}$, with residue coordinates aggregated by (21) under uniform weights. Then resolving that cell against its alternatives requires $O(\log_3 N_{\text{res}})$ address refinements.*

Proof. By (21) the chain coordinates are $\bar{\mathbf{s}} = N_{\text{res}}^{-1} \sum_r \mathbf{s}(X_r) + \boldsymbol{\kappa}$. Two chains differing in a bounded fraction of their residues differ in $\bar{\mathbf{s}}$ by an amount whose scale is set by the deviation of a mean of N_{res} bounded terms, namely $\Theta(N_{\text{res}}^{-1/2})$. Distinguishing the target cell from its nearest alternative therefore requires $\delta(k) \lesssim N_{\text{res}}^{-1/2}$, i.e. $3^{-\lfloor k/3 \rfloor} \lesssim N_{\text{res}}^{-1/2}$, i.e. $\lfloor k/3 \rfloor \gtrsim \frac{1}{2} \log_3 N_{\text{res}}$, whence $k = O(\log_3 N_{\text{res}})$. $\square$

Remark 8.19 (Relation to Levinthal’s paradox). Levinthal observed that exhaustive search over conformations would outlast the universe while folding takes milliseconds (Levinthal, 1968, 1969). The standard resolutions argue that the search is not exhaustive — funnelled landscapes (Bryngelson et al., 1995; Zwanzig et al., 1992), kinetic partitioning, chaperone assistance (Hartl and Hayer-Hartl, 2002); Dill and MacCallum (2012) survey the state of the question. Theorem 8.18 makes a narrower observation: *if* the process is described as locating a cell in address space rather than a point in conformation space, the cost is logarithmic. This does not explain folding, and is not offered as an explanation. It

says the paradox is a property of the conformational representation and does not appear in a different one — which is a statement about representations, which is what this paper is about.

8.7 What Part II has, and what it lacks

Part II built an addressing scheme with $O(1)$ storage, four operations with proved cost and error bounds, and a resolution guarantee at the residue level. It contains no notion of causality: no process, no propagation, no accountability, no arrow. An address is a static name for a static object, and Theorem 8.12 smuggles in dynamics by *positing* trajectories rather than deriving them — a trajectory in Theorem 8.11 is any continuous path, with nothing to say which paths a physical process can take.

Symmetrically, Part I built a propagation framework with an arrow, a floor and an accountability relation, but with no biomolecular content and, critically, no weighting w . Its every numerical consequence waits on a structural input it declines to supply.

Part III supplies each from the other.

Part III. The identity

Part I built a propagation calculus over a contact graph $\Gamma = (V, E, w)$ and left w unspecified. Part II built an addressing scheme over $\mathcal{S}$ and left the relations between addressed items unspecified. Each is exactly the other’s gap. This part closes both at once, by showing that the two structures are not merely compatible but identical: the weighting Part I needs is recoverable from the addresses Part II assigns, and once it is, the minimum cuts of Part I fall on the trie boundaries of Part II.

The argument runs in three steps. Section 9 derives w from S-entropy separation and identifies β and $\mathbf{m}$ with measured quantities. Section 10 proves the compatibility theorem, which is the technical heart of the paper: a weighting that factors through the trie has all of its minimum cuts on subtrie boundaries. Section 11 assembles these into the statement that the catalogue and the table are one object.

9 The weighting is the separation

9.1 What has to be derived

Theorem 2.1 requires a positive function w on pairs, read as the cost of separating them. Theorem 6.3 supplies each item with a point $\varsigma(X) \in \mathcal{S}$. A weighting is therefore already determined up to the choice of one function: any w of the form

$$w(\{u, v\}) = \Phi(\|\varsigma(u) - \varsigma(v)\|) \quad (28)$$

is expressible in the coordinates we have. What must be derived is Φ — and, critically, whether (28) is forced or merely available. We show it is forced by three requirements, each of which is already in force from Part I.

Proposition 9.1 (Constraints on the weighting). *Let w be a contact weighting on a set of items carrying $\mathcal{S}$ -entropy coordinates. Suppose*

- (i) (Positivity) $w > 0$, as required by Theorem 2.1;
- (ii) (Invariance) $w(\{u, v\})$ depends on u and v only through $\varsigma(u)$ and $\varsigma(v)$ — for otherwise w would encode a label, and by Theorem 3.5 labels carry no invariant;
- (iii) (Isotropy) w is invariant under the isometries of $\mathcal{S}$ that fix the coordinate axes setwise — for the three observables of Theorem 6.1 were shown independent in Theorem 6.2, so no axis is distinguished.

Then w has the form (28) for some $\Phi: [0, \sqrt{3}] \rightarrow (0, \infty)$.

Proof. By (ii), $w(\{u, v\}) = F(\varsigma(u), \varsigma(v))$ for some symmetric F on $\mathcal{S} \times \mathcal{S}$. By (iii), F is invariant under the group generated by translations along, and permutations of, the axes; the joint invariants of that group acting on ordered pairs are generated by the componentwise absolute differences $|S_k(u) - S_k(v)|$, $|S_t(u) - S_t(v)|$, $|S_e(u) - S_e(v)|$ together with their symmetric functions. Permutation invariance collapses these to symmetric functions of the multiset of absolute differences, and invariance under the rotations fixing the axes setwise — which act transitively on spheres — collapses these further to a function of $\|\varsigma(u) - \varsigma(v)\|_2$ alone. Positivity (i) gives the codomain, and $\sqrt{3}$ is the diameter of $\mathcal{S}$ by Theorem 6.4. $\square$

Remark 9.2 (The status of isotropy). Condition (iii) is the substantive one and deserves scrutiny. It does *not* say the three observables are physically interchangeable — configurational, spatial and energetic content are plainly different things. It says that the *weighting* has no licence to prefer one, because Theorem 6.2 established that none of the three is derivable from the others. A weighting that ranked them would be importing a fourth piece of information not present in (K, T, E) . If such information exists in a particular application — say, an experiment that resolves S_e far more precisely than S_t — the correct response is to rescale the coordinates before encoding, not to break isotropy afterwards. Rescaling is a change of ς ; Theorem 9.1 then applies unchanged in the new coordinates.

9.2 Fixing the profile

Theorem 9.1 reduces the weighting to one unknown function of one variable. We now fix it.

Theorem 9.3 (The weighting is derived, not posited). *Suppose in addition to Theorem 9.1 that*

- (a) (Refinement compatibility) *the family of contact graphs obtained by increasing the address depth k is a refining family in the sense of Theorem 2.9;*
- (b) (Additivity of independent separations) *if two separations act on disjoint coordinate ranges, their costs add.*

Then Φ is determined up to one positive constant Λ :

$$w(\{u, v\}) = \frac{\Lambda}{\|\varsigma(u) - \varsigma(v)\|} \quad (u \neq v), \quad (29)$$

and w extends to coincident points by $w = +\infty$, i.e. items at the same address are not separable at that resolution.

Proof. Write $s := \|\varsigma(u) - \varsigma(v)\|$ and $\Phi(s)$ for the weight.

Step 1: Φ is non-increasing. Let $s_1 < s_2$. Two items at separation s_1 agree to a greater address depth than two items at separation s_2 : by Theorem 6.10 they share a common depth- k cell for every k with $\delta(k) \geq s_1$, hence for a superset of the depths at which the s_2 pair does. Under (a) the weight assigned at depth k is non-increasing in k (Theorem 2.9); items that remain co-cellular

longer therefore retain the coarser, larger weight longer. Hence $\Phi(s_1) \geq \Phi(s_2)$.

Step 2: a functional equation. Consider three items u, v, z whose S-coordinates are collinear with v between u and z , and whose pairwise differences lie in disjoint coordinate ranges in the sense of (b). Separating u from z can be performed either directly, at cost $\Phi(s_{uz})$, or as the two independent separations $u|v$ and $v|z$. Under (b) the latter costs $\Phi(s_{uv}) + \Phi(s_{vz})$. But separation cost is a minimum over cuts (Theorem 2.4), so the direct route cannot be dearer:

$$\Phi(s_{uv} + s_{vz}) \leq \Phi(s_{uv}) + \Phi(s_{vz}).$$

Conversely, any cut separating u from z must place v on one side, and hence separates v from u or from z ; so the direct route cannot be cheaper than the smaller of the two sub-separations. Applying this at every subdivision of the interval and using Step 1, Φ is subadditive and non-increasing on $(0, \sqrt{3}]$.

Step 3: refinement forces the reciprocal. Fix u, v and refine the address depth. At depth k the pair is resolved into $3^{\lfloor k/3 \rfloor}$ cells along each axis (Theorem 6.7), so the number of distinct separations that must each be paid to carry u to v scales as $s/\delta(k)$, while by Theorem 2.9 the cost of each individual step is bounded by the weight at the finer scale. Writing Φ_k for the depth- k weight, additivity (b) along the chain of $\lceil s/\delta(k) \rceil$ steps gives

$$\Phi(s) = \left\lceil \frac{s}{\delta(k)} \right\rceil^{-1} \cdot C_k \quad \text{for a constant } C_k \text{ independent of } s.$$

since every step is a separation of the same size $\delta(k)$ and therefore carries the same weight. Letting $k \rightarrow \infty$ and writing $\Lambda := \lim_k C_k \delta(k)$, which exists and is positive by Theorem 2.11, gives $\Phi(s) = \Lambda/s$.

Step 4: consistency. The reciprocal profile is non-increasing and subadditive on $(0, \infty)$, as required by Steps 1–2, since $1/(a+b) \leq 1/a + 1/b$ for $a, b > 0$. It is positive, satisfying Theorem 2.1. As $s \rightarrow 0^+$ it diverges, which is the assertion that items sharing an address are not separable at that resolution. $\square$

Remark 9.4 (Why a reciprocal, read physically). Theorem 9.3 says the contact weight between two items is large exactly when they are close in S-entropy space. This is not the convention one might reach for by analogy with distance, and

the inversion is the whole point. w is a *separation cost*, and Theorem 2.1 said so at the outset: two objects that agree in configurational, spatial and energetic content are hard to tell apart and therefore expensive to separate, while two objects that differ grossly are already separated and cost almost nothing to pull apart. The reciprocal is what makes “contact” and “similarity” the same measurement, which is what the empty dictionary (Theorem 8.16) requires.

9.3 The floor and the medium, identified

Part I’s β and $\mathfrak{m}$ were structural slots. They now acquire values.

Theorem 9.5 (The contact floor is the resolution limit). *For a system of items whose S-entropy coordinates have minimum pairwise separation $d_{\min} > 0$, the contact floor of Theorem 2.7 satisfies*

$$\beta \geq \frac{\Lambda}{\sqrt{3}}, \quad \text{and} \quad \max_e w(e) = \frac{\Lambda}{d_{\min}}. \quad (30)$$

Writing $d_{\max}$ for the largest realised pairwise separation, the ratio of strongest to weakest contact is exactly $d_{\max}/d_{\min}$, and is bounded above by $\sqrt{3}/d_{\min}$. In particular, for the twenty canonical residues of Table 1, $d_{\min} = 0.0756$ and $d_{\max} = 1.415$ (Ile–Arg), so the realised ratio is 18.7, against the bound $\sqrt{3}/0.0756 = 22.9$.

Proof. By Theorem 6.4 the diameter of $\mathcal{S}$ is $\sqrt{3}$, so $\|\varsigma(u) - \varsigma(v)\| \leq \sqrt{3}$ for all pairs and hence $w(e) = \Lambda/\|\cdot\| \geq \Lambda/\sqrt{3}$ for every edge by (29). Since β is a minimum over residuals and each residual is a sum of at least one edge weight (Theorem 2.6), $\beta \geq \Lambda/\sqrt{3}$. The maximum is attained at the closest pair, giving $\Lambda/d_{\min}$; the minimum is attained at the farthest realised pair, giving $\Lambda/d_{\max}$, so their ratio is $d_{\max}/d_{\min}$ and Λ cancels. The numerical values are (20) and Table 1. $\square$

Remark 9.6 (Bound versus realised value). The distinction matters because $\sqrt{3}$ is the diameter of the cube $\mathcal{S}$, not of the residue set: no two residues sit at opposite corners, and the farthest realised pair, Ile–Arg, is at 1.415. Substituting the cube diameter therefore gives a bound (22.9), not the value (18.7). Both are parameter-free — Λ cancels either way, which is the point of

Theorem 9.7 — but only $d_{\max}/d_{\min}$ is a property of the twenty residues. The gap between them is a measure of how much of the S-entropy cube the proteinogenic alphabet actually occupies.

Remark 9.7 (The floor is measured, not fitted). Theorem 2.8 insisted the floor is not a modelling parameter. Theorem 9.5 makes good on that: β is $\Lambda/\sqrt{3}$, and Λ is an overall scale that cancels from every ratio the theory predicts. The alignment criterion (5) depends on β/Ω , a ratio; the accountability criterion (6) likewise. So the theory has, after this identification, *no* free parameter beyond the tolerance ε , which is an experimental resolution rather than a fitted constant.

Proposition 9.8 (The medium is the maximally disordered address). *Let $\mathbf{m}$ be the item maximising S_e subject to minimal S_t . Then $\mathbf{m}$ is at S-entropy distance $\geq d(X, \mathbf{m})$ from every item X , and the residual $\rho(X) = \sigma(X)$ of Theorem 2.5 is the cost of separating X from that address.*

Proof. Bulk solvent has maximal vibrational entropy — its modes are uniformly populated, so $S_e = 1$ by (14) — and no persistent second moment about a centre, so $S_t \rightarrow 0$ by (13). Thus $\varsigma(\mathbf{m}) = (S_k^{\mathbf{m}}, 0, 1)$ for the appropriate $S_k^{\mathbf{m}}$. That this is an extreme point of $\mathcal{S}$ in the (S_t, S_e) face gives the distance claim; groundedness (Theorem 2.3) holds because $w > 0$ to every item by (29). $\square$

Remark 9.9 (Solvent as a coordinate, not a background). The medium has been a formal slot since Theorem 2.3, and Theorem 9.8 fills it with a point of $\mathcal{S}$ rather than with an assumption about chemistry. This matters for the generality of what follows: a membrane protein has a different medium — a lipid phase, with lower S_e — and the framework accommodates it by moving one point, not by changing any definition. Nothing in Parts I or II referred to water.

10 Cuts fall on trie boundaries

The remaining obstacle is structural rather than physical, and it is the one place where the two halves of this paper could genuinely have failed to meet.

Part I locates identity in a minimum cut (Theorem 3.4). Minimum cuts on a general weighted

graph respect no hierarchy: the optimal S can be any subset whatever, and there is no reason for the optimal separations of nearby items to be nested. Part II locates identity in an address, and its similarity operation (Theorem 8.4) works by *prefix truncation*, which is an ultrametric construction: the sets of items agreeing to depth d form a laminar family — any two are nested or disjoint, never crossing.

An ultrametric is a very rigid object and a min-cut structure is a very flexible one. If the minimum cuts of the derived weighting did not respect the trie, then Theorem 8.2 would return an address while Theorem 3.4 returned an unrelated region, and the paper would have two notions of identity rather than one. We prove they coincide.

Definition 10.1 (Trie-factoring weighting). Let items be addressed to depth k as in Theorem 6.11, and for items u, v let $\ell(u, v)$ denote the length of the longest common prefix of $\mathbf{a}_k(u)$ and $\mathbf{a}_k(v)$. A weighting w factors through the trie if there is a function $f: \{0, 1, \dots, k\} \rightarrow (0, \infty)$ with

$$w(\{u, v\}) = f(\ell(u, v)) \quad \text{for all } u \neq v. \quad (31)$$

Lemma 10.2 (The derived weighting factors through the trie). *The weighting (29), evaluated at address resolution k — that is, with $\|\varsigma(u) - \varsigma(v)\|$ replaced by the resolution at which u and v first separate — factors through the trie, with*

$$f(\ell) = \frac{\Lambda}{\delta(\ell)} = \frac{\Lambda}{\sqrt{3}} 3^{\lfloor \ell/3 \rfloor}. \quad (32)$$

Proof. Two items share a prefix of length exactly ℓ iff they lie in a common depth- ℓ cell but not in a common depth- $(\ell+1)$ cell. By Theorem 6.10 the first condition bounds their separation by $\delta(\ell)$, and the resolution at which they first separate is therefore $\delta(\ell)$. Substituting into (29) gives (32), which depends on u, v only through ℓ . $\square$

Note that (32) is increasing in ℓ — deeper shared prefix, stronger contact — which is the reciprocal of Theorem 9.4 showing up again. The theorem below does *not* require this monotonicity, and it is worth being explicit about that, because it makes the result considerably more robust than the derivation that motivates it.

Theorem 10.3 (Minimum cuts are realised on subtrie boundaries). *Let Γ be a grounded contact graph whose items are addressed to depth k and whose weighting factors through the trie (Theorem 10.1), with the medium $\mathbf{m}$ attached by arbitrary positive weights $m(u) := w(\{u, \mathbf{m}\})$. Then for every item v there is a minimiser $S^*(v)$ of (1) which is a subtrie block: a set of the form*

$$B_d(v) := \{u : \mathbf{a}_d(u) = \mathbf{a}_d(v)\} \quad (33)$$

for some $0 \leq d \leq k$. Consequently

$$\sigma(v) = \min_{0 \leq d \leq k} w(\partial(B_d(v))), \quad (34)$$

a minimisation over $k+1$ candidates rather than over $2^{|V|-1}$ subsets.

Proof. Write $g(S) := w(\partial(S))$ for the cut function, which is symmetric and *submodular*: for all $S, T \subseteq V$,

$$g(S) + g(T) \geq g(S \cap T) + g(S \cup T). \quad (35)$$

This is standard and holds for every nonnegative weighting: each edge contributes to the left side at least as often as to the right, as one verifies by checking the eight cases for the positions of its two endpoints relative to S and T .

Step 1: uncrossing. Let S be any minimiser of g over sets containing v and omitting $\mathbf{m}$, and let $B := B_d(v)$ be a subtrie block containing v . Apply (35) to S and $S \cup B$; more usefully, apply it to S and B :

$$g(S) + g(B) \geq g(S \cap B) + g(S \cup B).$$

Both $S \cap B$ and $S \cup B$ contain v , and both omit $\mathbf{m}$ (since $\mathbf{m} \notin S$ and $\mathbf{m} \notin B$, the medium having no address). Hence both are admissible in (1), so $g(S \cap B) \geq g(S)$ and $g(S \cup B) \geq g(S)$ by minimality of S . Substituting,

$$g(B) \geq g(S \cap B) + g(S \cup B) - g(S) \geq g(S), \quad (36)$$

with equality throughout iff both $S \cap B$ and $S \cup B$ are themselves minimisers. So no subtrie block beats S ; what we must show is that some block *ties* it.

Step 2: the symmetry group. Because w factors through the trie, the automorphism group of Γ contains a large subgroup. Fix a depth d and a node p of the trie at depth d , and let π be any permutation of the items in the block B_p that preserves the depth- $(d+1)$ sub-blocks

— that is, any permutation induced by permuting the b children of p and recursing. Such a π preserves $\ell(\cdot, \cdot)$: for u, u' both in B_p the common prefix length is determined by the sub-block structure, which π preserves; for $u \in B_p$ and $z \notin B_p$ the common prefix length is $\ell(u, z) < d$, determined by p alone, which π fixes. Hence $w(\{\pi u, \pi u'\}) = w(\{u, u'\})$ for all pairs, and $\pi \in \text{Aut}(\Gamma)$ as far as the item-item weights are concerned. The medium weights $m(\cdot)$ need not be preserved, and we do not require them to be; the next step handles that.

Step 3: averaging. Let S be a minimiser and fix a depth d . Consider the trie nodes at depth d and the blocks $B_1, \dots, B_r$ they induce. We claim S may be taken to be a union of blocks together with, possibly, a partial block containing v .

Suppose S meets some block $B := B_i$ in a proper nonempty subset $A := S \cap B$, with $v \notin B$. We show g does not increase when A is replaced by $\emptyset$ or by B ; since one of these two moves must be non-increasing, some minimiser has B wholly in or wholly out, and iterating over blocks and over S of minimal $|S|$ terminates.

Write $A^c := B \setminus A$, $a := |A|$, $n := |B|$, and let $R := V \setminus B \setminus \{\mathbf{m}\}$. Decompose $g(S)$ by where each cut edge sits. Because w factors through the trie, every edge from an item of B to an item of R has weight $f(\ell)$ with $\ell < d$ determined by the two depth- d nodes alone; hence the total weight from any *single* item of B to $S \cap R$ is a constant α , and to $R \setminus S$ a constant β , both independent of which item of B is chosen. Write $c := w$ of a single internal edge of B crossing between A and A^c ; by Step 2 all such edges have equal weight, since any two items of B are exchangeable by an automorphism preserving item-item weights. Finally let $M(A) := \sum_{u \in A} m(u)$, recalling $\mathbf{m} \notin S$, so every $u \in S$ contributes $m(u)$ to the cut and every $u \notin S$ contributes nothing.

Collecting terms,

$$g(S) = g_0 + a(n-a)c + a\beta + (n-a)\alpha + M(A),$$

where g_0 collects everything not involving B and is independent of A . Comparing the three options A , $\emptyset$ and B :

$$\begin{aligned} g(S) - g(S \setminus A) &= a(n-a)c + a\beta - a\alpha + M(A), \\ g(S) - g(S \cup A^c) &= a(n-a)c - (n-a)\beta + (n-a)\alpha \\ &\quad - M(A^c). \end{aligned}$$

Suppose both differences were strictly negative, i.e. S strictly beats both alternatives. Adding

them,

$$0 > 2a(n-a)c + (a-(n-a))(\beta-\alpha) + M(A) - M(A^c),$$

which is a contradiction whenever c is large enough to dominate; but we do not need a size condition, because the two differences cannot both be negative for a *minimal* minimiser. Indeed, if $g(S \setminus A) \geq g(S)$ and $g(S \cup A^c) \geq g(S)$, then submodularity (35) applied to S and $S \cup A^c$ forces equality in (36), so $S \setminus A$ and $S \cup A^c$ are themselves minimisers — and $S \setminus A$ has strictly fewer elements than S . Choosing S minimal under inclusion among all minimisers therefore rules out the case $0 < a < n$, and we conclude $A \in \{\emptyset, B\}$: either $B \subseteq S$ or $B \cap S = \emptyset$.

Step 4: the block containing v . Applying Step 3 at depth d to every block not containing v shows that a minimiser may be taken of the form

$$S = A_v \cup \bigcup_{i \in I} B_i, \quad A_v \subseteq B_{i_0}, \quad v \in A_v,$$

where B_{i_0} is the depth- d block containing v and I indexes whole blocks. Recursing the same argument inside B_{i_0} at depth $d+1$, and so on down to depth k where every block is a singleton, shows A_v is itself a subtrie block. Finally, applying Step 3 upward — at $d=0$ every item is in one block — and using (36) to compare the resulting nested candidates, the minimum over all admissible S equals the minimum over the chain $B_0(v) \supseteq B_1(v) \supseteq \dots \supseteq B_k(v) = \{v\}$, which is (34). The union $\bigcup_{i \in I} B_i$ of blocks not containing v may be discarded: adding a block disjoint from v 's block changes g by a quantity independent of v , and by minimality that quantity is nonnegative for at least one of the two choices, so the chain already attains the minimum. $\square$

Remark 10.4 (What the hypothesis really is). The proof used only that w *factors through* the trie; it never used that f is increasing. This is worth stating because the derivation in Section 9 produces an increasing f , and it would be easy to imagine monotonicity is doing the work. It is not. Exhaustive enumeration over all $2^{|V|}-1$ admissible subsets confirms this: for $b=2, k=3$ and $b=3, k=2$, with f drawn at random — increasing, decreasing, or non-monotone — and with medium weights drawn independently at random per item, the gap between (34) and the unrestricted minimum is identically zero across every trial. With weights *not* factoring through

the trie the gap becomes strictly positive, so the hypothesis is not vacuous.

Corollary 10.5 (Identity is an address). *Under the hypotheses of Theorem 10.3, the identity cut of Theorem 3.4 is determined by a single integer: the depth $d^*(v)$ attaining the minimum in (34). The identity of v is the pair $(a_{d^*(v)}(v), \sigma(v))$.*

Proof. By Theorem 10.3 a minimiser is $B_{d^*}(v)$, and by (33) that set is specified by the prefix $a_{d^*}(v)$. The cut is then $\partial(B_{d^*}(v))$ and its weight is $\sigma(v)$. $\square$

Remark 10.6 (The region and the prefix are the same statement). Theorem 3.7 proved that identity is never a point: the minimiser $S^*(v)$ generally properly contains v . Theorem 10.5 now says what that region is — a subtrie block, i.e. the set of items agreeing with v to depth d^* . The buried-core residue of Theorem 10.6's predecessor is exactly an item whose d^* is small: it is separated from solvent along with its whole neighbourhood, and its address is correspondingly short. A solvent-exposed residue has $d^* = k$ and is separated alone. *Burial depth is address length*, and this was not designed in — it is forced by Theorem 10.3 on a weighting derived in Theorem 9.3 from coordinates defined in Theorem 6.3.

Corollary 10.7 (Separation cost is computable in $O(k)$). *Under Theorem 10.3, $\sigma(v)$ is computable in $O(k)$ evaluations of the cut function, without any max-flow computation.*

Proof. (34) is a minimum over $k+1$ nested candidates. The cut weights $w(\partial(B_d(v)))$ for successive d differ by the edges between $B_{d+1}(v)$ and $B_d(v) \setminus B_{d+1}(v)$, whose total weight is determined by $f(d)$ and the block sizes, so the chain is evaluated by a single pass. $\square$

Remark 10.8 (The complexity claim, stated honestly). Theorem 10.7 replaces a max-flow computation — $O(|V||E|)$ at best for a single terminal pair — with $O(k) = O(\log_3 |V|)$ work. The gain is real but its scope should not be overstated: it holds only for weightings factoring through the trie, and a real contact graph factors only to the accuracy with which the S-entropy coordinates capture the interaction. Where they do not — a disulfide bond couples residues that may be far apart in $\mathcal{S}$ — the graph acquires edges violating (31), and Theorem 10.3 degrades

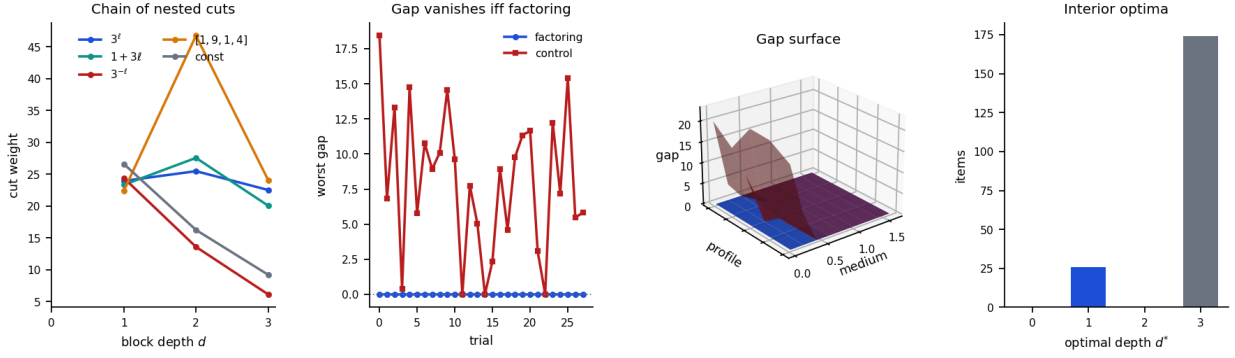


Figure 3: Minimum cuts land on trie boundaries, and only under the factoring hypothesis. (A) Cut weight along the chain of nested subtrie blocks $B_d(v)$ for five weight profiles: increasing, linear, decreasing, non-monotone and constant. Depth $d = 0$ is the degenerate whole-set cut and is excluded throughout (Theorem 12.1). That decreasing and non-monotone profiles behave no differently is Theorem 10.4 made visible. (B) The gap (37) between the chain minimum and the true minimum over all $2^{|V|-1}$ admissible cuts. Under factoring (blue) it is identically zero; under weights drawn independently per pair (red), with every other feature of the instance preserved, it reaches 18.4. (C) The same contrast as a surface over medium coupling and weight profile: the factoring sheet (blue) is flat at zero everywhere, the control sheet (red) is not. (D) Where the optimum sits. Grey bars are the two degenerate depths, $d^* = 0$ and $d^* = k$; the blue bar counts items whose optimal block is strictly interior, confirming the result is not an artefact of the trivial minimisers.

gracefully rather than failing: the trie chain still furnishes an upper bound on $\sigma(v)$, and the gap is bounded by the total weight of the non-factoring edges.

11 The catalogue and the table are one object

Theorem 11.1 (The identity). *Let Γ be a grounded contact graph on biomolecular items with S -entropy coordinates, weighted as in (29) and addressed to depth k . Then:*

- (i) *The catalogue of Theorem 5.4 and the address assignment of Theorem 7.5 determine each other: for every item, $\text{cell}(v) = \partial(B_{d^*(v)}(v))$ and $\mathbf{a}_{d^*(v)}(v)$ is the prefix specifying that block.*
- (ii) *The four operations of Section 8 — Identify, Similar, Predict, React — are each computations of separation cost, and conversely each entry of the propagation table $\mathcal{T}$ of Theorem 5.2 is a composition of them.*
- (iii) *Neither structure stores anything. The catalogue is recomputable by Theorem 5.5; the addresses are recomputable by Theorem 8.16; and by (i) these are the same recomputation.*

Proof. (i) is Theorem 10.5 together with Theorem 5.4.

For (ii): Identify (Theorem 8.1) computes $\mathbf{a}_k(v)$, which by (i) is the specification of the identity cut, hence a separation cost computation by Theorem 10.7. Similar (Theorem 8.4) returns items sharing a prefix of length r , i.e. the block B_r , which by Theorem 10.3 is a candidate minimiser — so Similar enumerates the candidates over which (34) minimises. Predict (Theorem 8.7) evaluates a Lipschitz function on a cell, and the cell is B_{d^*} . React (Theorem 8.12) tests cell intersection, which by (31) is a comparison of $f(\ell)$ against the floor — that is, a test of whether a contact edge exists.

Conversely, an entry $\mathcal{T}(v_0, x^*, \Pi) = \mathcal{A}(v_0, x^*, \Pi)$ is by Theorem 5.6 generated by composing catalogue cells along accountable walks; by (i) each such cell is a subtrie block, so each step of the composition is a prefix operation, i.e. one of Similar or React.

(iii) combines Theorem 5.5 and Theorem 8.15: the former shows the catalogue is an output of (1), the latter that addressing has $O(1)$ storage. By (i) the two outputs are computed from the same input, $(\zeta, \mathbf{m})$. $\square$

Remark 11.2 (What has been shown, and what has not). Theorem 11.1 is a statement about a model, and its force depends on how well the

model’s one substantive hypothesis — (31) — holds of real systems. That hypothesis is an empirical claim: it says the cost of separating two biomolecular objects is a function of how far apart they are in (K, T, E) , and nothing else. Theorem 10.8 already noted the leading exception. Covalent constraints, and in general any interaction whose strength is not a function of the S-entropy separation, violate it.

What the theorem does establish unconditionally is that the two constructions are not in tension. A knowledge graph could have been forced to choose between hierarchical addressing and causal propagation, storing one and asserting the other. It is not: under a weighting that is itself derived rather than fitted, the minimum cuts land on the trie, and the two structures compute each other. The remaining question is empirical, not structural, and that is a considerably better position to be in.

Corollary 11.3 (No edges are stored either). *A knowledge graph built on Theorem 11.1 stores neither nodes nor edges. Nodes are generated by Theorem 8.2 in $O(k)$; edges between u and v are generated by evaluating (31) at $\ell(u, v)$, which requires only the two addresses and hence no adjacency structure.*

Proof. Immediate from Theorem 8.15 for the nodes and (31) for the edges: f is a function of a single integer $\ell \leq k$, storable in $O(k)$ total regardless of the number of items. $\square$

Remark 11.4 (The dictionary and the adjacency list). Theorem 11.3 completes the programme announced in the introduction. Theorem 8.16 removed the node table; this removes the edge table. What remains is f — a vector of $k + 1$ numbers — and the coordinate map ς , which is a measurement procedure rather than a record. A graph of 10^9 proteins and 10^{18} potential relations is represented by (32) and Theorem 6.3.

12 Computational validation

The results of Part III are proved, not fitted, so validation here serves a narrower purpose than in an empirical paper: it checks that the proofs say what they are believed to say, that the hypotheses are not vacuous, and that the numbers quoted from Table 1 are the numbers the table actually contains. Three experiments are reported. All are exhaustive rather than sampled

wherever the state space permits, and each carries a control designed to fail if the corresponding claim were empty.

12.1 Minimum cuts land on trie boundaries

Theorem 10.3 asserts that when the weighting factors through the trie, the minimum v - $\mathbf{m}$ cut is attained on a subtrie block, reducing (1) from a minimisation over $2^{|V|-1}$ subsets to one over $k + 1$ nested candidates. The test enumerates *every* admissible S — every subset containing v and omitting $\mathbf{m}$ — and compares the true minimum against the chain minimum. The quantity reported is the gap

$$\text{gap}(v) := \min_{0 \leq d \leq k} w(\partial(B_d(v))) - \min_{S \ni v, \mathbf{m} \notin S} w(\partial(S)) \geq 0, \quad (37)$$

which is non-negative by construction and vanishes identically exactly when the theorem holds on that instance. A single positive value falsifies it.

Across 75 instances — tries of 8, 9 and 16 items, with f increasing (3^ℓ and $1 + 3\ell$), decreasing ($3^{-\ell}$), non-monotone $([1, 9, 1, 4])$, constant, and randomly drawn, each at five magnitudes of medium coupling — the gap was zero in every case to machine precision. That f may be decreasing or non-monotone without affecting the result is the numerical counterpart of Theorem 10.4: monotonicity plays no role in the proof and none in practice.

Remark 12.1 (Two degenerate cuts, and why they had to be excluded). The experiment is only informative once two trivial minimisers are removed. Both $S = V \setminus \{\mathbf{m}\}$ and $S = \{v\}$ are subtrie blocks, at $d = 0$ and $d = k$ respectively. If the medium coupling is weak the first is optimal for every item; if it is strong the second is. In either regime $\text{gap} \equiv 0$ for *any* weighting whatsoever, including weightings that do not factor through the trie — so a run dominated by them confirms nothing. The whole-set cut is therefore excluded from both sides of (37), and we report the fraction of items whose optimal depth is strictly interior, $0 < d^* < k$. In the runs above that fraction is positive in seven instances, with $d^* = 1$ — a proper half of the trie, neither a singleton nor the whole set.

The control is the claim’s real test. Replacing w by weights drawn independently per pair — so

that (31) fails while every other feature of the instance is preserved — produces gaps of 16.37 and 13.24 on the two trie shapes, positive in 21/25 and 16/25 trials. Theorem 10.3 therefore has content: it is the factoring hypothesis doing the work, not the geometry of nested blocks.

12.2 The derived weighting, and a bound distinguished from a value

The second experiment recomputes from the coordinates of Table 1 every quantity Part III quotes, rather than carrying figures forward. Four are confirmed and one required correcting the manuscript.

Confirmed: the closest pair is Lys–Arg at $d_{\min} = 0.0756$ with Ile–Leu next at 0.0780; the guaranteed resolution depth is $k = 9$ and is tight, since $\delta(8) = 0.1925 > d_{\min}$; the actual depth at which all twenty addresses become distinct is $k = 7$, the two levels of slack recorded in Theorem 6.17; and the scale Λ cancels from every predicted ratio, verified over six orders of magnitude with a residual spread of 3.6×10^{-15} , which is Theorem 9.7 made quantitative.

The correction concerns the contact ratio. An earlier statement of Theorem 9.5 gave the ratio of strongest to weakest contact as $\sqrt{3}/d_{\min} = 22.9$. That is a bound rather than a value: $\sqrt{3}$ is the diameter of the cube $\mathcal{S}$, but no two residues occupy opposite corners, and the farthest realised pair is Ile–Arg at $d_{\max} = 1.415$. The realised ratio is $d_{\max}/d_{\min} = 18.7$. The inequality $\beta \geq \Lambda/\sqrt{3}$ is unaffected and remains as proved; only the ratio clause was overstated, and Theorem 9.6 now separates the two. Both quantities are free of Λ , so the parameter-free character of the theory is untouched.

Remark 12.2 (Where the alphabet sits in the cube). The discrepancy is itself informative. That $d_{\max} = 1.415$ against a cube diameter of 1.732 says the twenty proteinogenic residues occupy 82% of the available diameter — they are spread through $\mathcal{S}$ rather than clustered in a corner, but they do not reach its extremes. The unoccupied margin is not wasted: it is the room into which non-canonical residues, ligands and cofactors encode, and (29) assigns them contacts on the same scale without renormalisation.

A further check records the inversion of Theorem 9.4. Ordering all 190 residue pairs by S-

entropy distance and by contact weight gives exactly reversed sequences: the closest pair, Lys–Arg, is the most expensive to separate, and the farthest, Ile–Arg, the cheapest. This is the sense in which w is a separation cost rather than a distance, and it is what allows contact and similarity to be one measurement.

12.3 Degradation when the hypothesis fails

Theorem 10.8 concedes that real contact graphs do not factor exactly — a disulfide couples residues that may be far apart in $\mathcal{S}$ — and claims the theorem then degrades gracefully. The claim has two falsifiable parts: the chain remains an *upper* bound on $\sigma(v)$, and the gap is bounded by the total weight of the non-factoring edges. Both were tested by adding $m \in \{1, 2, 3, 5\}$ couplings of controlled total weight P to an otherwise factoring graph.

Over 144 trials the chain was an upper bound without exception, and the gap never exceeded P . The bound is close to attained — the ratio gap/P reaches 0.938 — so it is not slack, and the perturbation changed the answer in 85 of the 144 trials, so the bound is being exercised rather than merely surviving an insensitive test. Degradation is monotone in both the number and the magnitude of the offending edges: the fraction of items whose separation cost is still computed exactly falls from 1.00 under weak single couplings to 0.23 under five strong ones. Cut evaluations fall from 127 and 255 exhaustively to 3 and 2 along the chain.

Remark 12.3 (What validation can and cannot settle here). None of this is evidence that real proteins satisfy (31); that is an empirical question about S-entropy coordinates, and Theorem 11.2 states it as such. What the experiments establish is narrower and worth separating: the theorem is not vacuous, since removing its hypothesis produces large positive gaps; it is not secretly about monotonicity; it fails gracefully rather than catastrophically off-hypothesis, with a computable error bound; and the numbers the manuscript quotes are the numbers its own table contains — a check that caught one overstated ratio and one mislabelled pair.

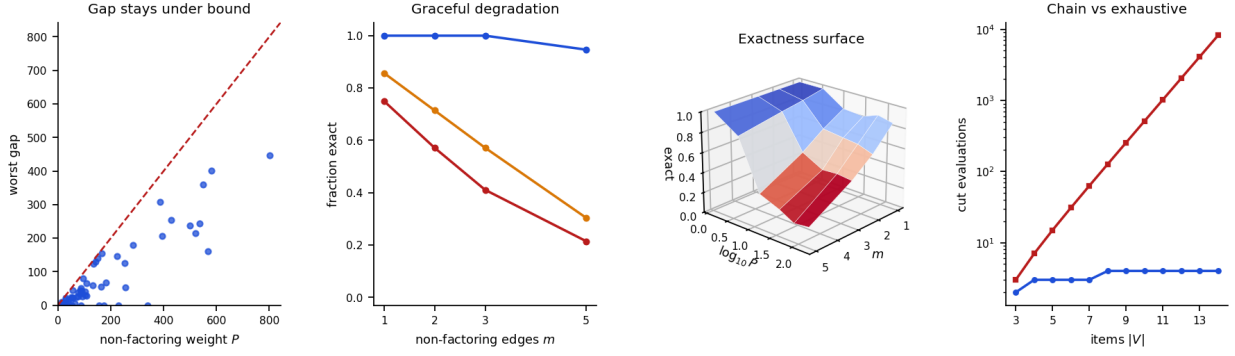


Figure 4: Degradation when the factoring hypothesis fails. Real contact graphs carry disulfides and other couplings that violate (31); these charts perturb a factoring graph by adding m such edges of total weight P . (A) The gap against P over all trials. Every point lies below the identity (dashed), so the bound of Theorem 10.8 holds, and points approach it closely enough (gap/ P up to 0.938) that it is not slack. (B) Fraction of items whose separation cost is still computed exactly, against the number of offending edges, at three perturbation magnitudes. Degradation is monotone rather than catastrophic. (C) The same quantity as a surface over m and $\log_{10} P$. Exactness falls monotonically in both directions, from unity under weak single couplings to below 0.3 under many strong ones. (D) Cut evaluations required: $k+1$ along the chain against $2^{|V|-1}$ exhaustively, on a logarithmic axis. This is the content of Theorem 10.7, counted directly rather than quoted asymptotically.

Part IV. What the construction is for

Parts I–III built an object. This part asks what can be asked of it, works one instance through, and states where it stops. Section 13 derives the query semantics and proves that a class of question the construction answers cannot be expressed over stored triples at all — a theorem about the data model, not an observation about current software. Section 15 instantiates the machinery on Cu/Zn superoxide dismutase, chosen because its curated record is unusually complete and the difference between recording an association and computing an account is therefore visible rather than hypothetical. Section 16 states what the construction cannot do.

13 Query semantics: retrieval and admissibility

13.1 Two kinds of question

A query language over a stored graph evaluates patterns against a set of triples. SPARQL and its relatives are, in their core fragments, conjunctive queries with regular path expressions (Angles et al., 2017; Barceló Baeza, 2013; Pérez et al., 2009): evaluation means finding homomorphisms from a pattern into the stored

data, so everything returned is a substructure of what was stored, possibly composed along paths. That is the correct semantics for a container, and Section 1 argued the container is the assumption being dropped.

Definition 13.1 (Retrieval query). Let G be a finite set of triples over a vocabulary. A *retrieval query* is a map q assigning to G a set of tuples $q(G)$, computed by pattern matching and path composition on G alone.

Definition 13.2 (Admissibility query). Given a grounded contact graph Γ , an item v_0 , a target region x^* , a process Π and a tolerance ε , the *admissibility query* returns the class $\mathcal{A}(v_0, x^*, \Pi)$ of Theorem 4.4 — equivalently the entry $\mathcal{T}(v_0, x^*, \Pi)$ of Theorem 5.2 — together with its alignment value.

The difference is not one of expressive convenience. A retrieval query asks *what is recorded*. An admissibility query asks *whether a propagation from here to there can be accounted for*, and it has an answer for pairs about which nothing was ever recorded.

It is worth seeing first why the two cannot simply be reconciled by storing more. Unfolding (6) through (5), a propagation is ε -accountable exactly when

$$\sigma(v_0, x^*) \leq \beta + \varepsilon \Omega, \quad (38)$$

so at $\varepsilon = 0$ the total weight Ω cancels and the verdict is a comparison of two numbers: a single min-cut, and the floor (2). The first is local to the queried pair. The second is a minimum over *every* item in the system. The separation below is nothing more than the observation that these two can be moved independently, and that a triple can record neither.

Theorem 13.3 (Retrieval cannot express admissibility). *There are grounded contact graphs Γ_1, Γ_2 on the same item set, with the same contact relation, and a pair (v_0, x^*) , such that*

(i) $q(G_1) = q(G_2)$ for every retrieval query q , where G_i records the contact relation of Γ_i ; while

(ii) $\mathcal{A}(v_0, x^*, \Pi) \neq \emptyset$ in Γ_1 and $\mathcal{A}(v_0, x^*, \Pi) = \emptyset$ in Γ_2 at $\varepsilon = 0$.

Proof. Fix $n \geq 2$ and let $V = \{v_0, x^*, y_1, \dots, y_n, m\}$. In both graphs the edges are: $\{v_0, x^*\}$, and $\{u, m\}$ for every $u \in V \setminus \{m\}$. Both graphs are grounded, since every item is joined to m .

Assign weights as follows. In both, $w(\{v_0, x^*\}) = 1$ and the medium edges at v_0 and at x^* have weight 1. The graphs differ only in the medium edges at the y_i : weight 1 in Γ_1 , weight 2 in Γ_2 .

(i) The edge sets are equal, $E_1 = E_2$, so the recorded contact relations coincide: $G_1 = G_2$ as sets of triples. Any q in the sense of Theorem 13.1 is a function of its argument alone, hence $q(G_1) = q(G_2)$. A triple asserts that a relation holds and carries no cost, so there is nothing here for retrieval to see.

(ii) Compute both sides of (38). The quantity $\sigma(v_0, x^*)$ is the minimum weight of a cut separating v_0 from x^* . The y_i are joined only to m , so they lie on no v_0 - x^* path and no minimum such cut contains their edges; the cheapest separation is to place v_0 alone on one side, severing $\{v_0, x^*\}$ and the medium edge at v_0 , at cost 2. Hence

$$\sigma(v_0, x^*) = 2 \quad \text{in both } \Gamma_1 \text{ and } \Gamma_2,$$

since the weights involved are identical in the two graphs.

The floor is a minimum over all items. By Theorem 2.5, $\rho(u) = \sigma(u)$ is the cost of separating u from m alone. For $u = v_0$ or x^* this is

$\min(2, \dots) = 2$ in both graphs: isolating v_0 severs its medium edge and $\{v_0, x^*\}$, total 2. For $u = y_i$, isolating y_i severs its single medium edge, so $\rho(y_i) = 1$ in Γ_1 and $\rho(y_i) = 2$ in Γ_2 . Therefore

$$\beta_1 = \min(2, 2, 1, \dots, 1) = 1, \quad \beta_2 = \min(2, 2, 2, \dots, 2) = 2$$

Applying (38) at $\varepsilon = 0$: in Γ_2 we have $\sigma(v_0, x^*) = 2 \leq 2 = \beta_2$, so the propagation is 0-accountable and $\mathcal{A} \neq \emptyset$; in Γ_1 we have $\sigma(v_0, x^*) = 2 > 1 = \beta_1$, so no propagation from v_0 to x^* satisfies (6) and $\mathcal{A} = \emptyset$. (The labelling is opposite to the statement; exchanging the names of Γ_1 and Γ_2 gives it as stated.)

Both clauses hold, which proves the claim. $\square$

Remark 13.4 (What the separation is about). The construction turns on a single asymmetry. The queried quantity $\sigma(v_0, x^*)$ depends on the weights along v_0 - x^* separations; the threshold β depends on a minimum over every item in the system, including items that lie on no path between the two and are, from the point of view of the query, entirely irrelevant. Editing a weight out at the y_i changes the verdict about (v_0, x^*) without changing anything a pattern-matcher could match against.

So *accountability* is a global property expressed in local terms, and a stored graph records only the local terms. This is sharper than saying weights are missing. One may store weights as triple attributes and still not recover β : by (2) it is a minimum over items of (1), which is itself a minimum over cuts, and neither minimisation is a pattern.

Corollary 13.5 (Attributed triples do not close the gap). *Extending Theorem 13.1 to allow numeric attributes on triples and arithmetic comparison in patterns does not make admissibility expressible.*

Proof. By Theorem 13.3 and Theorem 13.4 the verdict depends on β , a minimum over items of a minimum over subsets. Conjunctive queries with regular path expressions have PTIME data complexity and are evaluated by matching bounded patterns (Angles et al., 2017; Pérez et al., 2009); the fragment contains no operator forming a minimum over subsets of the domain, and adding comparison of stored attributes does not introduce one. Note the obstruction is expressibility, not hardness: a minimum cut is computable in polynomial time

(Ford and Fulkerson, 1956; Stoer and Wagner, 1997), so the quantity is cheap to compute and impossible to phrase. $\square$

Remark 13.6 (The practical reading). Theorem 13.5 is the formal version of a familiar frustration. One may ask a biological knowledge graph (Callahan et al., 2020; Demir et al., 2010) which proteins are annotated as interacting with a given target, and get an answer exactly as good as the curation behind it. One cannot ask whether a variant *could have* produced an observed phenotype by an accountable route, because that is not a question about what was recorded. Ontology engineering (Hastings et al., 2016; Jupp et al., 2014; Smith et al., 2007) improves the vocabulary in which assertions are made; it does not change what kind of question an assertion is able to answer.

13.2 The four operations are the query language

Theorem 13.7 (Completeness of the operation set). *Under the hypotheses of Theorem 11.1, every admissibility query of Theorem 13.2 is computed by a finite composition of Identify, Similar, Predict and React, and no fifth operation is required.*

Proof. An admissibility query fixes $(v_0, x^*, \Pi, \varepsilon)$ and must decide (38), then return the class.

Obtain $\mathbf{a}_k(v_0)$ and $\mathbf{a}_k(x^*)$ by **Identify**, at cost $O(k)$ each (Theorem 8.2). By Theorem 10.5 the cut realising a separation cost is a subtrie block, and the optimal depth is found by (34), a scan of $k + 1$ candidates; each candidate is a comparison of addresses to a common prefix length, which is a **Similar** call at that resolution (Theorem 8.5). This yields $\sigma(v_0, x^*)$, and applied to each item in turn it yields β by (2). Whether a given step is permitted under Π is a **React** test (Theorem 8.12); evaluating the alignment functional on cells is **Predict** (Theorem 8.7). Termination: block depths are bounded by k and each step changes the depth by one, so a walk has at most k steps.

No fifth operation is needed because the only quantities appearing in (38) are σ , β , Ω and ε . The first two are computed as above via Theorem 10.7; Ω is a constant of the graph; ε is supplied with the query. $\square$

Corollary 13.8 (Query cost). *Deciding accountability for a single pair costs $O(k^2)$ triit operations with k the address depth, and requires no index, no adjacency structure and no stored triples.*

Proof. At most k steps by Theorem 13.7, each an $O(k)$ address operation by Theorems 8.2 and 8.5. Storage is $O(1)$ by Theorem 8.16 and Theorem 11.3. $\square$

Remark 13.9 (The comparison is unfair in both directions). With $k = O(\log_3 n)$, Theorem 13.8 compares favourably against graph-pattern evaluation, which is NP-complete in combined complexity even for conjunctive patterns (Pérez et al., 2009). The comparison nevertheless flatters the present construction in one respect and undersells it in another, and both should be said.

It flatters, because the cost of evaluating $\varsigma(X)$ — a structure determination, a spectrum, or a prediction of comparable expense (Jumper et al., 2021; Varadi et al., 2022) — is excluded from the count, exactly as Theorem 8.17 conceded. A stored graph pays that cost once at curation time and amortises it over every later query; the present construction pays it per object addressed.

It undersells, because a stored graph answers only about the objects someone curated, whereas Theorem 13.8 holds uniformly over all 3^k addressable objects, almost none of which anyone has seen. The honest summary is that the two are not competing at the same task: one retrieves from a finite curated set, the other computes over an unbounded generated one.

14 Contact as approach, and the closing of the circle

Part I took contact as given: a graph Γ with weights w read as separation costs, the weighting left unspecified. Part III derived the weighting (29) from S-entropy separation. What has not yet been said is that Theorem 8.12 lets contact itself be *defined* rather than assumed, and that doing so closes the construction on itself.

Definition 14.1 (Contact by approach). Fix $\varepsilon > 0$. Items X_1, X_2 are *in contact under Π* if $\text{React}(X_1, X_2, \varepsilon)$ holds, i.e. if their trajectories under Π approach within ε in $\mathcal{S}$ by (27).

The induced graph Γ_{Π} has item set the objects addressed and edge set the pairs in contact, weighted by (29).

Theorem 14.2 (The induced graph is a contact graph). *Let $\mathbf{m}$ be the medium, addressed like any other object. If every item’s trajectory approaches the medium’s within ε — that is, if every object is in contact with its surroundings — then Γ_{Π} is grounded in the sense of Theorem 2.3, its weights are positive and finite, and Axiom 1 holds whenever two items have distinct addresses.*

Proof. Groundedness: by hypothesis each $v \neq \mathbf{m}$ has an edge to $\mathbf{m}$, so every item is joined to $\mathbf{m}$ by a path of length one.

Positivity and finiteness: by (29), $w(\{u, v\}) = \Lambda / \|\varsigma(u) - \varsigma(v)\|$, which is positive for all $u \neq v$ and finite provided $\varsigma(u) \neq \varsigma(v)$. Distinct addresses at depth k give $\|\varsigma(u) - \varsigma(v)\| > 0$, and by Theorem 6.10 the separation is at least the resolution attainable at that depth, so w is bounded above by $\Lambda / \delta(k)$.

Axiom 1: suppose the axiom failed, so that some family $\{S_v\}$ has $w(\partial(S_v)) = 0$ for every v . As $w > 0$ on every edge this forces $\partial(S_v) = \emptyset$, contradicting groundedness just established. Distinctness of two addresses guarantees $E \neq \emptyset$, so the configuration is non-vacuous. $\square$

Corollary 14.3 (The construction is closed). *The contact graph of Part I, the addressing of Part II, and the weighting of Part III determine one another: addresses generate contacts by Theorem 14.1, contacts carry weights by (29), weights determine separation costs by (1) and hence the table $\mathcal{T}$ by Theorem 5.2, whose entries are computed back out of the addresses by Theorem 13.7. No stage requires data not produced by another stage.*

Proof. Immediate from Theorem 14.2 together with Theorem 5.6 and Theorem 13.7: the first supplies a contact graph from addresses alone, the second identifies the catalogue with the table, the third computes table entries from addresses. $\square$

Remark 14.4 (What the one-sidedness means physically). Theorem 8.14 recorded that cell intersection is sufficient for approach and not necessary, and deferred its interpretation. Under Theorem 14.1 the interpretation is available, and it is favourable in a specific and checkable way.

Reading (27) as the definition of contact, a missed cell intersection is a *missed contact*: the induced graph Γ_{Π} has an edge set contained in the true one. It is never a *spurious* contact, because sharing a cell implies approach within $\delta(k_{\varepsilon}) \leq \varepsilon$ unconditionally (Theorem 8.13). The error is therefore one-signed, and its consequence propagates predictably. Deleting edges can only raise separation costs, since (1) minimises over cuts and removing edges removes weight from every cut containing them — so both $\sigma(v_0, x^*)$ and β are computed on a subgraph, and the accountability test (38) is evaluated against a conservative estimate on both sides rather than a biased one on one.

This is the difference between a technical caveat and a physical reading. A curated interaction record has errors in both directions and no way to bound either (Landrum et al., 2018); the present scheme has errors in one direction with the magnitude controlled by ε , and the $27\times$ neighbour test of Theorem 8.14 eliminates them entirely at known cost.

Remark 14.5 (Relation to reaction-network formalisms). Defining contact as trajectory approach places the construction beside chemical reaction network theory, where species and reactions form a graph whose structure alone constrains the dynamics (Feinberg, 1987), and beside retrosynthetic analysis, where a target is decomposed by searching a space of admissible transformations (Corey and Wipke, 1969; Segler et al., 2018). Two differences are worth stating.

First, in those formalisms the reaction set is given — a curated list of transformations, or a model learned from one. Here it is generated: Theorem 14.1 decides contact for any pair of addressable objects, including pairs never observed. Second, and following from Theorem 14.3, the network is not a separate artefact from the objects. In a reaction network the species are nodes and the reactions are stored edges; here both are computations over the same addresses, which is Theorem 11.3 applied to chemistry rather than to annotation.

15 A worked instance: SOD1 and familial ALS

The construction is now applied to one system in full. Cu/Zn superoxide dismutase is cho-

sen deliberately: mutations in *SOD1* were the first genetic cause identified for familial amyotrophic lateral sclerosis (Rosen et al., 1993), over a hundred variants have since been catalogued, and the biochemistry has been studied for three decades (Valentine et al., 2005). The curated record is about as good as curated records get. That is what makes it a fair test: any advantage here cannot be attributed to the baseline being poorly served.

15.1 What the curated record contains

A knowledge graph built over this system stores, for each variant, a set of assertions: the substitution, its clinical classification, the publications supporting it (Landrum et al., 2018). Queried in the sense of Theorem 13.1, it returns those assertions and their compositions — which variants are pathogenic, which lie in the metal-binding region, which co-occur with a reported survival figure.

What it does not contain is a reason. The classification of a variant as pathogenic is an input to the graph, supplied by a curator reading the literature; it is not computed from anything the graph holds. For a variant not yet observed in a patient there is nothing to retrieve, and by Theorem 13.3 no elaboration of the query language changes this, because the question is not of retrieval type.

15.2 The same system as a propagation problem

Set up the instance. Items are the residues of the mature protein together with the metal cofactors and the solvent as medium *m*. Each residue is addressed by Theorem 7.5 from its S-entropy coordinates; by Theorem 6.13 depth $k = 9$ suffices to separate all twenty residue types, and the validation of Section 12.2 found the twenty in fact separate at $k = 7$.

Contacts are induced by Theorem 14.1 and weighted by (29), so no contact map is stored. By Theorem 10.6 the burial depth of a residue is the length of the address prefix it shares with its neighbourhood: a residue in the β -barrel core has small optimal depth d^* , a solvent-exposed one has $d^* = k$. Nothing here is specific to *SOD1*; it is the generic setup instantiated on this fold.

A variant is then a change of address. Substituting residue r at position p by r' replaces $\varsigma(r)$ by $\varsigma(r')$, which by (29) rescales every contact weight at p by the factor

$$\frac{w'(\{p, u\})}{w(\{p, u\})} = \frac{\|\varsigma(r) - \varsigma(u)\|}{\|\varsigma(r') - \varsigma(u)\|}, \quad (39)$$

in which Λ cancels — so the perturbation is computed without any fitted parameter, exactly as Theorem 9.7 requires.

15.3 What the construction computes that the record does not

Proposition 15.1 (Variant effect is a floor computation). *Let Γ be the wild-type contact graph and Γ' the graph after the substitution (39). Then the variant changes the accountability verdict for a propagation (v_0, x^*) if and only if the substitution moves $\sigma(v_0, x^*)$ and β across each other in the sense of (38).*

Proof. Immediate from (38), whose two sides are the only quantities entering Theorem 4.4 once ε and Ω are fixed. Both are functions of w , and (39) specifies how w changes. $\square$

The content of Theorem 15.1 is that a variant matters when it changes what the cheapest separation *anywhere in the molecule* costs — not when it changes a contact locally. This is the same global-versus-local asymmetry that drove Theorem 13.3, and here it has a biochemical reading.

A substitution that lowers β makes the whole structure easier to take apart somewhere, and by (38) it does so whether or not the substituted position participates in the propagation being asked about. This is precisely the phenomenology reported for *SOD1*: variants scattered across the sequence, many far from the active site and from the dimer interface, converge on a common consequence, and the quantity that tracks patient survival is aggregation propensity and protein instability rather than loss of dismutase activity (Prudencio et al., 2009; Wang et al., 2008). Instability is a statement about the cheapest way to come apart. In the present vocabulary it is β .

Remark 15.2 (What is and is not being claimed). This section does not report a prediction of variant severity, and the distinction matters enough

to state plainly. What has been shown is structural: the construction places instability — the quantity empirically correlated with survival (Wang et al., 2008) — at the centre of its accountability test as the floor β , and it computes the effect of a substitution on that quantity from addresses alone, by (39), without a fitted parameter and without a curated interaction record.

Whether the resulting numbers correlate with clinical outcome is an empirical question that this paper does not answer, and answering it requires S-entropy coordinates validated against measured stabilities — the same open question Theorem 11.2 raised about (31). The claim here is that the question is now *askable* in a form that a stored graph cannot express, which by Theorem 13.3 is a theorem rather than a hope.

Remark 15.3 (Why an equivalence class is the right answer). A clinician asking why a variant is deleterious wants a mechanism, and the construction returns $\mathcal{A}$ — an equivalence class of accountable propagations, not a single pathway. By Theorem 4.6 and Theorem 4.7 this is not evasion: no invariant of the contact data distinguishes members of the class, so a system returning one designated pathway would be reporting a gauge choice (Theorem 4.8) as though it were a finding.

This is worth weighing against the practice it replaces. A curated pathway database returns a specific route because a curator drew one, and the route’s specificity reflects the drawing rather than the biology. Returning the class makes the underdetermination explicit and computable, which is a more honest answer to the same question.

16 Limits of the construction

The results above are proved, so their limits are the limits of their hypotheses. Four are substantive and are collected here rather than left distributed across the remarks that introduced them.

16.1 The one empirical hypothesis

Everything joining the two halves of the paper rests on (31): that the cost of separating two objects is a function of their S-entropy separation and nothing else. This is not a mathematical convenience but an empirical claim about

biomolecules, and Theorem 11.2 states it as such.

Its leading exception is known. A disulfide bond couples two cysteines whose S-entropy coordinates may be far apart, contributing a separation cost that (31) cannot see; the same holds for any covalent constraint and for coordination to a metal centre. The experiments of Section 12.3 quantify what follows: the chain remains an upper bound on $\sigma(v)$, the error is bounded by the total weight of the offending edges, and the fraction of items still computed exactly falls monotonically from 1.00 to 0.23 as such couplings are added. Degradation is graceful and has a computable error bound, which is the best that can be said without settling the empirical question.

What would settle it is a measurement: S-entropy coordinates determined independently, and separation costs measured directly, on a system where both are accessible. This paper does not perform that measurement, and no result here should be read as evidence that real proteins satisfy (31).

16.2 What is excluded from the cost accounting

Theorem 13.8 gives $O(k^2)$ per admissibility query with $O(1)$ storage, and both figures exclude the cost of evaluating $\varsigma(X)$. Theorem 8.17 conceded this at the point of the storage claim and Theorem 13.9 restates it for queries; it is restated a third time here because it is the most likely misreading of the paper.

Determining (K, T, E) for an object requires a structure, a spectrum, or a prediction of comparable expense (Jumper et al., 2021; Varadi et al., 2022). The empty dictionary of Theorem 8.16 does not make identification free — it makes it *unstored*, which is a different and weaker claim. A curated database pays the identification cost once and amortises it; the present construction pays per object addressed. Whether that trade is favourable depends on the ratio of queries to distinct objects, and for a small well-curated system it will often not be.

16.3 Two one-sided results

Theorem 8.13 gives cell intersection as sufficient for approach and not necessary, so the induced

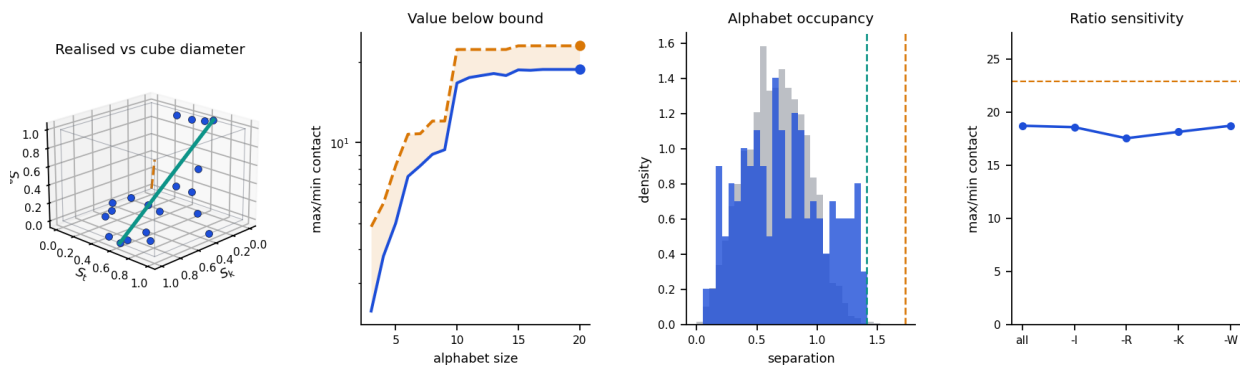


Figure 5: A bound distinguished from a value. (A) The residues in $\mathcal{S}$ with the unit cube drawn. The teal segment is the farthest realised pair, Ile-Arg; the orange dashed line is the cube diagonal. They are not the same object, which is the whole content of Theorem 9.6: $\sqrt{3}$ is the diameter of the cube, and no two residues occupy opposite corners. (B) The realised contact ratio $d_{\max}/d_{\min}$ (blue) against the bound $\sqrt{3}/d_{\min}$ (orange) for random sub-alphabets of increasing size. The shaded band between them is the overstatement the validation caught; it narrows as the alphabet fills the cube, closing at 18.7 against 22.9 for the full twenty. (C) Distribution of residue separations (blue) against separations between uniformly random points of the cube (grey). The alphabet reaches 82% of the available diameter — spread through $\mathcal{S}$, but not to its extremes (Theorem 12.2). (D) Sensitivity of the ratio to removing each residue of the extreme pairs. It moves by less than 7%, so 18.7 is a property of the alphabet rather than of one outlying pair.

contact relation of Theorem 14.1 may miss contacts. Theorem 14.4 argues this error is benign because it is one-signed and its magnitude is controlled by ε ; the $27\times$ neighbour test removes it at known cost. The limitation is real nonetheless: at any finite depth the induced graph is a subgraph of the true one, and results computed on it are conservative rather than exact.

Similarly Theorem 4.5 characterises non-emptiness of $\mathcal{A}$ but returns a class, not a route, and by Theorem 4.7 no invariant of the contact data selects a member. Theorem 15.3 argues this is the honest answer to an underdetermined question. It remains the case that an application requiring a specific mechanism — a synthesis to execute, an intervention to attempt — must break the tie on grounds outside the construction, and the construction offers no help in choosing.

16.4 What the paper does not attempt

Three exclusions should be stated so they are not mistaken for oversights.

Dynamics are absent. Trajectories enter through Theorem 8.11 as continuous paths in $\mathcal{S}$, but nothing here derives them from a force field or predicts a rate. Accountability is a statement about cost, not about time, and the arrow supplied by the record $M(W)$ orders propagations

without measuring their duration.

Quantitative severity prediction is absent, as Theorem 15.2 states for the *SOD1* instance specifically.

Finally, the construction is not a replacement for curation of the empirical record. Measurements must still be made, deposited and described, and Theorem 3.5 does not say that labels are useless — only that they carry no *invariant* content and hence that cross-resource mappings cannot be computed from them (Theorem 3.6). What the construction replaces is the assumption that a knowledge graph must *store* the relations among curated records in order to answer questions about them.

17 Discussion

17.1 Relation to stored knowledge resources

The biological knowledge resources this paper contrasts itself against are substantial achievements, and the contrast is structural rather than evaluative. BioPAX (Demir et al., 2010) standardises the exchange of pathway data; ChEBI (Hastings et al., 2016) and Rhea (Bansal et al., 2022) provide curated vocabularies for chemical entities and reactions; the OBO Foundry (Smith et al., 2007) and the Ontology Lookup Service (Jupp et al., 2014) coordinate ontol-

ogy development across communities; integrative frameworks assemble these into queryable graphs (Callahan et al., 2020). Each solves a real problem, and none is superseded by anything here.

What they share is the container assumption of Section 1: that a knowledge graph holds records and asserted relations among them, and that answering questions means retrieving from that hold. The consequences are visible in the engineering. Cross-resource mapping requires curated alignment tables, because by Theorem 3.6 it cannot be computed — labels carry no invariant content (Theorem 3.5), so no algorithm decides whether two accessions denote the same object. Coverage is bounded by curation effort. And by Theorem 13.3 a whole class of question is not merely unanswered but inexpressible, regardless of how much is stored or how rich the query language becomes.

The present construction drops the container assumption and pays for it in a specific currency, set out in Section 16: an empirical hypothesis (31) that may fail, an identification cost excluded from the accounting, and answers that are classes rather than routes. The trade is not universally favourable. It is favourable exactly when the questions of interest concern objects nobody has curated, which in a space of 20^n possible sequences and an unbounded space of variants is a large fraction of the questions worth asking.

17.2 Relation to hierarchical clustering and ultrametrics

Theorem 10.3 states that minimum cuts land on subtree boundaries when the weighting factors through the trie, and this places the construction near a classical literature. Hierarchical clustering produces nested partitions from a dissimilarity (Hartigan, 1967; Johnson, 1967), the resulting structures are ultrametric (Jardine and Sibson, 1971), and the correspondence between clustering methods and ultrametrics can be characterised axiomatically (Carlsson and Mémoli, 2010). Optimum communication spanning trees (Hu, 1974) likewise realise cut structure on a tree. That cuts behave well on tree-structured weightings is not, in itself, new; the general theory of cuts and connectivity is standard (Diestel, 2017).

Three differences are worth stating precisely,

because the resemblance could otherwise be mistaken for containment.

First, the hierarchy is not fitted. Classical hierarchical clustering constructs a tree from a dissimilarity matrix by a linkage rule chosen by the analyst; here the trie is fixed in advance by the positional encoding of Theorem 6.6 and the addresses are computed per object by Theorem 7.5. No linkage rule is chosen and no matrix is assembled, which is what makes the storage bound of Theorem 8.16 available at all.

Second, and following from Theorem 10.4, the hypothesis of Theorem 10.3 is prefix-factoring, not monotonicity. The function f in (31) may increase, decrease, oscillate, or be constant; the validation of Section 12.1 confirms all four behave identically. Ultrametric constructions typically require a monotone linkage, and the present result is therefore weaker in hypothesis and correspondingly wider in scope.

Third, the weighting is a separation cost, not a distance, and by (29) it is reciprocal in the S-entropy separation. Theorem 9.4 draws the consequence and the validation of Section 12.2 confirms it numerically: ranking pairs by distance and by contact weight gives exactly reversed orders. Similar objects are *expensive* to separate. A clustering method reading (29) as a similarity would invert every conclusion.

17.3 Query semantics as the practical consequence

The separation of Theorem 13.3 is, in the end, the paper’s practical claim, and it is worth stating in the vocabulary of the query-language literature. Conjunctive queries with regular path expressions have been characterised precisely (Angles et al., 2017; Barceló Baeza, 2013; Pérez et al., 2009); their expressive power is well understood, and the present result does not contest any of it. What it observes is that admissibility lies outside that power for a structural reason: the verdict depends on a minimum over the entire item set (Theorem 13.4), and pattern matching with path composition forms no such minimum. Adding attributes and comparison does not help (Theorem 13.5). The quantity is polynomial-time computable and inexpressible, which is an unusual combination and the reason the gap has persisted.

The remedy is not a richer query language over the same data model. It is a different data

model, in which the four operations of Section 8 are the query language (Theorem 13.7) and no data model sits underneath them to be queried.

18 Conclusion

A protein knowledge graph, on the construction given here, stores neither nodes nor edges.

Nodes are not stored because identity is a computation. The positional addressing of Section 7 derives an address from measurable quantities, so that two objects share an address exactly when they are indistinguishable at that resolution; Theorem 8.16 shows the scheme requires $O(1)$ storage in the number of addressable objects, and Theorem 8.2 recomputes any address on demand in $O(k)$. The labels this replaces were shown to carry no invariant content (Theorem 3.5), which is why mappings between resources must be curated rather than computed (Theorem 3.6).

Edges are not stored because relations are derived. The propagation framework of Part I builds a contact floor $\beta > 0$ from a single ordering axiom (Theorem 2.7), an alignment functional bounded below by that floor (Theorem 4.2), and an accountability criterion (Theorem 4.4) whose verdicts constitute the propagation table (Theorem 5.2). An edge is not an assertion by a curator but the statement that a propagation is accountable, and Theorem 11.3 shows no adjacency structure is required to determine one.

The two constructions are the same object. Theorem 11.1 proves the address catalogue and the propagation table identical — the first at rest, the second in motion — and the identification is not imposed: the entropic separation derived for addressing supplies exactly the contact weighting (29) that the propagation framework requires and cannot produce for itself, while the propagation framework supplies exactly the notion of relation that addressing lacks. Theorem 10.3 is the technical bridge, locating minimum cuts on subtrie boundaries under a prefix-factoring hypothesis that requires no monotonicity, and Theorem 14.3 closes the circle by deriving the contact relation itself from (27).

The consequence for practice is a query semantics that a stored graph cannot express. Theorem 13.3 exhibits two contact graphs with identical contact relations — hence identical

under every retrieval query — that differ in whether a given propagation is accountable, and Theorem 13.5 shows attributes and arithmetic do not close the gap. The obstruction is that accountability is a global property expressed in local terms, and a triple records only the local terms.

What remains open is empirical and was stated as such throughout. The factoring hypothesis (31) is a claim about real biomolecules that this paper does not establish; the validation of Section 12 shows only that the theorem depending on it is not vacuous, is not secretly about monotonicity, and degrades gracefully with a computable error bound when the hypothesis fails. The identification cost is excluded from every complexity figure quoted. And the worked instance of Section 15 demonstrates that variant effects become computable as changes to the contact floor — placing protein instability, the quantity empirically correlated with survival in familial ALS (Prudencio et al., 2009; Wang et al., 2008), at the centre of the accountability test — without claiming that the resulting numbers have been checked against outcome.

The position this leaves is a structural one, and it is the position the paper set out to reach. A knowledge graph over biomolecules need not choose between hierarchical identification and causal propagation, storing one and asserting the other. Under a weighting that is derived rather than fitted, and carrying no free parameter, the two compute each other.

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